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Neuro-adaptive control design for underactuated quadcopter drone: A fractional-order super-twisting sliding mode approach

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ABSTRACT

Unmanned aerial vehicles (UAVs), particularly quadcopters, pose significant control challenges due to their nonlinear dynamics, underactuated structure, and susceptibility to external disturbances such as wind gusts and sensor noise. To address these issues, this paper proposes a neuro-adaptive fractional-order super-twisting sliding mode control (FOST-SMC) framework that integrates a fractional-order super-twisting differentiator (FOST-D) and a deep residual neural network with attention mechanisms (DRNAM). The FOST-SMC guarantees finite-time convergence and reduces chattering with robustness even in the presence of matched uncertainites and disturbances. Adaptivity is enabled even for highly nonlinear input channels by DRNAM, which also reduces the need for accurate system models. Moreover, FOST-D performs well in providing accurate real-time velocity estimation in a noisy environment, thereby enhancing state feedback and overall system stability. Lyapunov-based analysis rigorously guarantees finite-time convergence, and high-fidelity MATLAB/Simulink results show that the proposed control scheme achieves better trajectory tracking and robustness than traditional methods. This overall strategy ensures robust performance in controlling quadcopter UAVs in an unpredictable, changing environment.

1. Introduction

Unmanned Aerial Vehicles (UAVs), in particular quadcopters, have attracted considerable attention owing to their wide range of applications in surveillance, aerial mapping, disaster management, and autonomous delivery [1]. On the other side, they present tough control challenges due to complex nonlinear dynamics, under-actuated nature, and the influence of external disturbances. For achieving robust and accurate trajectory tracking in the presence of uncertainties due to aerodynamic effects, change of payload, actuator faults, and sensor noise, advanced control methodologies are indeed required [2].

Traditional control approaches, including Proportional-Integral-Derivative (PID) control, Linear Quadratic Regulator (LQR), and Linear Quadratic Gaussian (LQG), have been widely adopted for the control of quadcopter systems because they are simple and easy to implement. Each method minimizes a cost function to realize some desired metrics, such as minimum energy consumption or tracking error accounted for so far in this research work [3,4]. However, most of these methods face their limit of achieving stability and performance since their control design relies on a linearized model and thus cannot present the full nonlinear nature of quadcopters. This growing demand fosters the emergence of more advanced nonlinear and adaptive control. Passivity-based control (PBC) has also been used for the stabilization of quadcopters. Energy properties of such a system have been tapped through the PBC methods to realize controllers that ensure performance and stability. The procedures have been used successfully in quadcopter systems and noted to have additional robustness and stability [5,6].

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Sliding mode control (SMC) is one of the most popular nonlinear control methods which applies robustness in quadcopter UAV systems, especially when uncertainties are present [7]. The approach provides stability and precise tracking by forcing system states onto a given sliding manifold. Under sliding mode control, the closed-loop dynamics exhibit strong robustness to matched parameter variations and external disturbances, though performance may still be influenced by unmatched uncertainties-a property particularly important to cope with real-world conditions. Intrinsic robustness in SMC makes the technique very ideal for quadcopter systems that will tend to be subjected to external disturbances, including wind gusts, payload fluctuations, and sensor noises. An integrator-supported sliding mode control method is constructed for quadcopter dynamics robust stabilization under uncertainty conditions [8,9]. By obliterating the reaching phase, the method forces robustness at the very beginning, resulting in sensational closed-loop performance improvements [10]. However, it should be noted that the integral sliding mode control method does not decrease the system order, i.e., the parameters and state of the whole system are all involved in sliding mode. The property may impact the control performance, particularly on parametric variation sensitive systems and dynamic perturbation. To alleviate high frequency chattering issue of SMC, higher-order sliding mode approaches such as the Super-Twisting Sliding Mode Control (ST-SMC) have been proposed [11]. ST-SMC improves convergence speed and reduces chattering compared to classical SMC, but it still requires accurate system modeling and struggles under strong nonlinearities and noisy measurements.

Fuzzy logic control is another technique that has been used very widely in quadcopter systems. Fuzzy logic controllers (FLCs) use linguistic rules to address uncertainties and nonlinearities, which are most appropriately addressed in uncertain and complex environments. Fuzzy logic blended with sliding mode control, or Fuzzy Sliding Mode Control (FSMC), has been proven to minimize chattering and enhance robustness [12,13]. This hybrid method integrates the advantages of both methods to offer a flexible, adaptive control method for quadcopter systems. Event-triggered control is a next-generation approach that minimizes the computational burden and communication loads in control systems. In event-triggered control strategies, control updates are done only when necessary; hence, efficiency is optimized and resource usage is minimized. Event-triggered control approaches have been used for quadcopter systems and demonstrated superior improvements in performance and resource usage [14,15]. For instance, event-triggered MPC has been formulated to reduce the computational burden with optimal control performance. Hybrid Control approaches have also been investigated for quadcopter systems, wherein two or more control approaches are hybridized. For instance, combining sliding mode control with adaptive control or fuzzy logic control can benefit from the strengths of both and result in enhanced robustness and performance [16,17]. Switched Control techniques, which alternate among various control laws depending on system behavior, have also been investigated in the case of quadcopter systems. These techniques are very effective for controlling multiple operating mode systems or systems with large dynamics variations. Switched control techniques have been shown to offer robust and adaptive control performance in quadcopter systems [18,19].

In addition, backstepping control has also been of concern due to its systematic design process and capacity to manage nonlinear systems with unmatched uncertainties. Through recursive control law design for subsystems, backstepping control provides stability and tracking performance to the system. When merged with sliding mode control, backstepping methods have proven to improve robustness and eliminate chattering in quadcopter systems [20,21]. It is especially effective in managing the underactuated dynamics of quadcopters.

To further enhance output feedback control methodologies' robustness against uncertainties as well as disturbances from outside sources, observer-based methods have grown to become highlighted ones. Observer-based techniques estimate such uncertain disturbances using observers and then controllers get reconfigured for maintaining the stability of the system, as outlined in previous literature [22–24]. Those observer-based methodologies have proven efficient in forcing the state variables of a vehicle towards desired state variables. The techniques of observer-based sliding mode control proposed in [25] demonstrate chattering suppression while improving robustness to uncertainties. In Levant's work [26], a high-order robust exact differentiator with constant gains was designed within the framework of an output-feedback controller to estimate the derivative states, assuming that the system position is bounded. Unfortunately, the execution of this controller may be questionable in practice as systems are often subject to high levels of noise and/or fast dynamic fluctuations. Another efficient approach that enhances the robust performance of control systems is the Disturbance Observer-Based Control (DOBC). DOBC methods estimate and counteract disturbances in real-time to provide steady, accurate control performance. DOBC methods have been implemented successfully on quadcopter systems and provided enhanced disturbance rejection and track accuracy [27,28]. Nonlinear disturbance observers, for example, have been employed for external disturbance estimation and cancellation to achieve accurate control of quadcopter dynamics. Several research papers have incorporated the use of neural networks in controller design to enhance robustness in approximating near unknown nonlinearities and disturbances [29,30]. When combined with observers, these methods based on neural networks have been found to be extremely efficient in enhancing robustness of control systems. By approximating complex terms successfully, these methods bring about more robust and precise control systems.

Reinforcement Learning (RL) is now becoming increasingly popular for controlling quadcopter systems. RL controllers learn the best policies via experience while exploring the environment, hence with very high adaptability to evolving conditions. Deep Reinforcement Learning (DRL) and Q-learning methods have been used for quadcopter systems, which exhibited deep performance and adaptability improvements [31,32]. In addition to neural networks, adaptive control methods have extensively been studied to cope with uncertainties in quadcopters. Adaptive control methods tune controller parameters online depending on system performance, providing robustness against model uncertainty and external disturbances. For example, Model Reference Adaptive Control (MRAC) and Lyapunov-based adaptive control have been successfully implemented in quadcopter systems, providing enhanced tracking performance and disturbance rejection [33,34]. They are especially appropriate for use when system dynamics are not yet well established or undergo extreme fluctuations. Robust Control techniques, e.g., H-infinity control and μ-synthesis, have been attempted for quadcopters. These methods design controllers offering the best performance under uncertainties and disturbances. Robust control techniques have been found to achieve high robust stability guarantees and disturbance rejection in quadcopter systems [35,36]. For example, H-infinity control has been applied to design controllers, leading to the least disturbance effect on system performance with assured robust stability and performance.

Predictive control approaches, such as Model Predictive Control (MPC), have also been proposed for quadcopter systems. MPC applies to finite-horizon optimal control optimization of the control input in the presence of system constraints and, therefore, finds applications in those situations where stringent trajectory tracking and avoidance of obstacles are required. Robustness and adaptivity features were implemented in MPC during recent advancements to respond against uncertainties and disturbances effectively [37,38]. For instance, effective MPC methods have been developed to account for bounded model uncertainties and disturbances to provide stable and accurate control performance under stringent constraints. Nonlinear Model Predictive Control (NMPC) is another sophisticated control method that has been implemented for quadcopter systems. NMPC predicts the system's future states based on its nonlinear model and determines the optimal control inputs by solving a constrained finite-time optimal control problem. The approach is known to yield very good tracking results and is robust to uncertainties [39,40].

For instance, NMPC has been used to track accurate trajectories in the presence of obstacles and external disturbances.

Another promising direction for investigation is fractional-order control methods, which generalize integer-order control methods based on fractional calculus. Fractional-Order PID (FOPID) and Fractional-Order Sliding Mode Control (FOSMC) controllers offer greater controller parameter freedom to achieve better performance in handling nonlinear dynamics and system uncertainties [41,42].

Although sliding mode and observer-based control strategies for the quadcopter UAV have been studied recently, many problems remain open. Classical SMCs induce chattering, which has adverse effects on the quality of the output [43], high-frequency actuator vibration, and premature aging. Higher-order sliding mode controllers and observer-based schemes were developed to alleviate this problem, but these methods rely on strict knowledge of the system dynamics and input channels, which is an unrealistic assumption in practical applications with uncertainties and disturbances. Neural network-based adaptive control schemes, albeit effective for nonlinear approximation, do not account for fractional-order dynamics or fast time-varying disturbances. Estimators based on high-gain observers or robust differentiators are very sensitive to noise and perform poorly in the presence of strong disturbances. Furthermore, fractional-order calculus applications in sliding mode control are still in their infancy, especially for quadcopter systems that require greater robustness and flexibility. Finally, most adaptive and observer-based schemes do not exhibit finite-time convergence, which is crucial for drone operations in real time, as they require fast and precise control. The drawbacks of these techniques motivate us to propose a novel framework based on fractional-order calculus, NN adaptation, and robust differentiators, which provides a unified solution for quadcopter UAV control in the presence of unknown perturbations.

1. **Novel NA-FOSTSMC framework:** A Neuro-Adaptive Fractional-Order Super-Twisting Sliding Mode Control (NA-FOSTSMC) framework is proposed for trajectory tracking of under-actuated quadcopters under unknown external disturbances. Unlike PID and classical SMC controllers that suffer from poor robustness and chattering, and standard ST-SMC approaches that rely heavily on accurate models, the proposed framework integrates fractional calculus to achieve finite-time convergence and enhanced disturbance rejection.
2. **Integration of DRNAM for adaptability:** A Deep Residual Neural Network with Attention Mechanisms (DRNAM) is incorporated to estimate nonlinear drift terms and input channels. Compared to simple feed-forward neural networks used in prior adaptive control studies, DRNAM enables deeper feature extraction and dynamic prioritization of relevant states, thereby improving adaptability and reducing dependence on precise system models.
3. **Development of FOST-SMD differentiator:** A Fractional-Order Super-Twisting Sliding Mode Differentiator (FOST-SMD) is proposed to estimate velocity states based on the noisy position measurements. Traditional differentiators are sensitive to noise and amplify it, whereas the fractional-order-based approach provides dependable feedback in the presence of uncertainties and enhances the robustness of the solution when applied to UAVs.

Besides the methodological contributions, the proposed approach has some practical merits. The use of fractional calculus and super-twisting sliding mode is employed to ensure finite-time convergence of the closed-loop system and to eliminate chattering in the presence of matched uncertainties and external disturbances. In addition, a strict Lyapunov stability analysis is used to formally prove the finite-time stability of the entire control system. The effectiveness and practicality of the proposed approach are further validated through high-fidelity MATLAB/Simulink simulations, which demonstrate superior trajectory tracking accuracy, enhanced robustness, and improved control efficiency when compared with conventional control strategies.

The rest of this paper is organized as follows. Section 2 provides the problem formulation of the quadcopter UAV. The newly proposed control design is explained in Section 3. The comparison of performance between the proposed scheme and previous results is provided in Section 4. Finally, concluding remarks are summarized in Section 5.

2. Problem formulation

The quadcopter's rotational motion is defined by roll φ, pitch ϑ, and yaw ψ, while its translational motion consists of lateral position (x, y) and altitude z. The quadcopter's dynamics are expressed as follows:

$$\left.\begin{aligned}
\ddot{\varphi} &= \frac{I_R\dot{\vartheta}\bar{\omega}}{I_x} + \frac{(I_y - I_z)}{I_x}\dot{\vartheta}\dot{\psi} + \frac{bl}{I_x}U_2 \\
\ddot{\vartheta} &= \frac{-I_R\dot{\varphi}\bar{\omega}}{I_y} + \frac{(I_z - I_x)}{I_y}\dot{\varphi}\dot{\psi} + \frac{bl}{I_y}U_3 \\
\ddot{\psi} &= \frac{(I_x - I_y)}{I_z}\dot{\varphi}\dot{\vartheta} + \frac{d}{I_z}U_4 \\
\ddot{x} &= (\sin\psi\sin\varphi + \cos\psi\sin\vartheta\cos\varphi)\frac{b}{m}U_1 \\
\ddot{y} &= (\sin\psi\sin\vartheta\cos\varphi - \cos\psi\sin\varphi)\frac{b}{m}U_1 \\
\ddot{z} &= (\cos\vartheta\cos\varphi)\frac{b}{m}U_1 - g
\end{aligned}\right\} \tag{1}$$

where b is the thrust coefficient, d is the drag factor, and I_R is the rotor inertia, l represents the arm length, while $\bar{\omega} = \omega_1 + \omega_3 - \omega_2 - \omega_4$ accounts for aerodynamic disturbances, ω_i $(i = 1, 2, 3, 4)$ represents the angular speed of each motor, the control inputs are $U_2 = -\omega_2^2 + \omega_4^2$, $U_3 = \omega_1^2 - \omega_3^2$, $U_4 = -\omega_1^2 + \omega_2^2 - \omega_3^2 + \omega_4^2$, the total thrust input for translational motion is given by U_1, and the angular speeds of the motors in terms of control inputs are $\Omega_1 = \sqrt{\frac{U_1}{4} + \frac{U_3}{2} - \frac{U_4}{4}}$, $\Omega_2 = \sqrt{\frac{U_1}{4} - \frac{U_2}{2} + \frac{U_4}{4}}$, $\Omega_3 = \sqrt{\frac{U_1}{4} - \frac{U_3}{2} - \frac{U_4}{4}}$, $\Omega_4 = \sqrt{\frac{U_1}{4} + \frac{U_2}{2} + \frac{U_4}{4}}$.

Since underactuated systems have fewer control inputs than degrees of freedom, the nonlinear system is transformed into an equivalent regular form, resulting in the state-space representation:

$$\left.\begin{aligned}
\dot{\psi}_1 &= \psi_2; \quad \dot{\psi}_2 = a_5\vartheta_2\varphi_2 + b_3U_4 \\
\dot{z}_1 &= z_2; \quad \dot{z}_2 = -g + (\cos\vartheta_1\cos\varphi_1)b_0U_1
\end{aligned}\right\} : \text{Fully-Actuated} \tag{2}$$

$$\left.\begin{aligned}
\dot{\hbar}_1 &= \hbar_2; \quad \dot{\hbar}_2 = gB_\hbar, \\
\dot{\vartheta}_1 &= \vartheta_2; \quad \dot{\vartheta}_2 = a_3\psi_2\varphi_2 + a_4\bar{\omega}\,\varphi_2 + b_2U_3 \\
\dot{\rho}_1 &= \rho_2; \quad \dot{\rho}_2 = gB_\rho, \\
\dot{\varphi}_1 &= \varphi_2; \quad \dot{\varphi}_2 = a_1\psi_2\vartheta_2 + a_2\bar{\omega}\,\vartheta_2 + b_1U_2
\end{aligned}\right\} : \text{Underactuated} \tag{3}$$

where $B_\hbar = \frac{\sin\psi\sin\varphi + \cos\psi\sin\vartheta\cos\varphi}{\cos\vartheta\cos\varphi}$, $B_\rho = \frac{\sin\psi\sin\vartheta\cos\varphi - \cos\psi\sin\varphi}{\cos\vartheta\cos\varphi}$, and $b_0 = \frac{b}{m}$, $a_1 = \frac{I_y - I_z}{I_x}$, $a_2 = \frac{I_R}{I_x}$, $a_3 = \frac{I_z - I_x}{I_y}$, $a_4 = -\frac{I_R}{I_y}$, $a_5 = \frac{I_x - I_y}{I_z}$.

Assumption 1. The matched type disturbances Δ_z, Δ_ψ, Δ_ϕ, Δ_θ satisfy

$$\|\Delta_z\| \le \lambda_z, \quad \|\Delta_\psi\| \le \lambda_\psi, \quad \|\Delta_\phi\| \le \lambda_\phi, \quad \|\Delta_\theta\| \le \lambda_\theta,$$

where λ_z, λ_ψ, λ_ϕ, λ_θ are positive constants.

In the next section, the control design is discussed.

3. Control design

This section presents a smart and efficient finite-time control strategy for quadcopter systems, achieved through the integration of a NA-FOSTSMC with a FOST-SMD. The proposed framework is designed to ensure accurate trajectory tracking, strong robustness against external disturbances, and adaptability under model uncertainties. Essentially, in the proposed algorithm, the system states are shown to converge to the desired trajectories in finite time, and the chattering effect is efficiently attenuated by fractional calculus and super-twisting mechanisms. This design can preserve robustness with respect to matched uncertainties

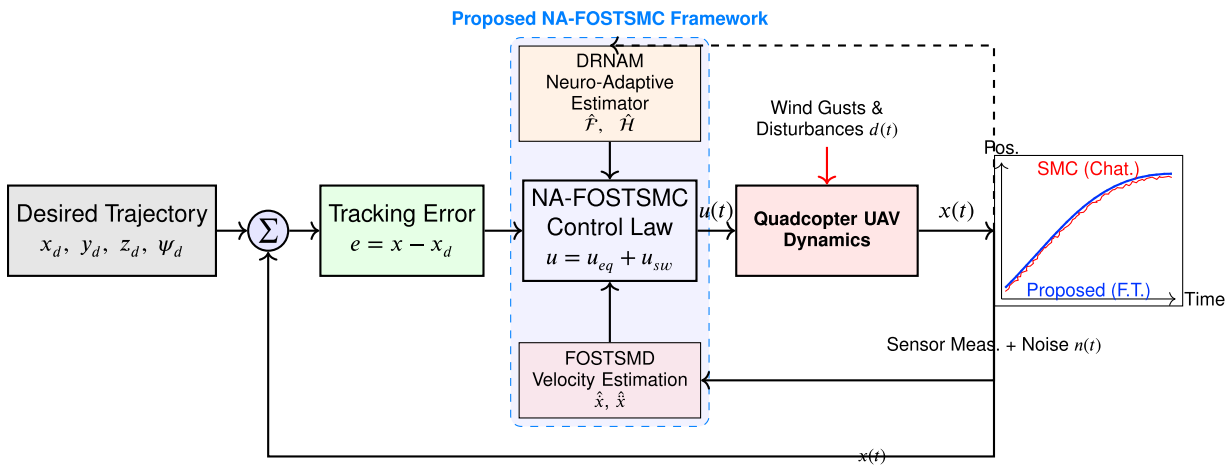

Fig. 1. Detailed complete block diagram of the proposed NA-FOSTSMC framework for quadcopter trajectory tracking, integrating fractional-order sliding surface, DRNAM estimator, FOSTSMD differentiator, UAV dynamics, disturbance injection, and performance comparison..

while improving steady-state performance. The DRNAM is then added to represent the extremely nonlinear input channels resulting from a system normal-form transformation, thereby enhancing flexibility. Based on residual learning and attention mechanisms, the DRNAM achieves high control accuracy without requiring accurate model information of the quadcopter, even in dynamic and uncertain environments. Furthermore, the FOST-SMD is applied to estimate linear and angular velocities in real time with high accuracy, despite being affected by sensor noise and modeling uncertainties, making it the core of the proposed nonlinear observer and providing trustworthy state feedback. The integration of the differentiator significantly enhances closed-loop stability and overall control performance. By embedding these components into a unified architecture, the proposed control strategy achieves finite-time convergence, adaptive capability, and precise trajectory tracking. The complete control structure is illustrated in Fig. 1, where all developed modules are seamlessly integrated for quadcopter UAV applications. The control objective is to ensure accurate tracking of the desired trajectories x_d, y_d, z_d, ψ_d while stabilizing the roll and pitch angles.

The following subsections explain how to design both control protocols and give a full stability analysis.

3.1. Fractional order super-twisting sliding mode control design

The control structure is divided into two parts: one for the fully-actuated subsystem and another for the under-actuated subsystem. Finite-time trajectory tracking for both the subsystems will be provided with matched uncertainties. First, the under-actuated subsystem should be considered, followed by the fully-actuated subsystem.

3.1.1. Control design of the under-actuated subsystem

The under-actuated subsystems' dynamics, given as ($\rho - \varphi$ and $\hbar - \vartheta$), are controlled by Eqn. (3). As previously mentioned, both subsystems possess similar dynamic structures. Hence, for brevity, the ($\rho - \varphi$) subsystem will be used to provide a representative control design.

The primary objective is to drive the system states to their desired trajectories within finite time. To achieve this, we introduce a fractional-order super-twisting sliding manifold defined as:

$$s_\rho = \dot{\varepsilon}_\rho + \alpha_\rho \varepsilon_\rho + \beta_\rho D^{\gamma_\rho} \varepsilon_\rho, \tag{4}$$

where $\varepsilon_\rho = \rho_1(t) - \rho_d(t)$ denotes the tracking error, $\alpha_\rho > \beta_\rho > 0$ are positive design constants, γ_ρ is the fractional order ($0 < \gamma_\rho < 1$), and D^{γ_ρ} represents the Caputo fractional derivative operator.

The inclusion of the fractional derivative term $D^{\gamma_\rho}\varepsilon_\rho$ introduces a memory effect into the control law. This nonlocal aspect enables the controller to consider the entire history of the tracking error, resulting in greater robustness and better convergence in a certain sense [44,45]. The Caputo derivative can be expressed as:

$$D^{\gamma_\rho}\varepsilon_\rho(t) = \frac{1}{\Gamma(1-\gamma_\rho)} \frac{d}{dt} \int_0^t \frac{\varepsilon_\rho(\tau)}{(t-\tau)^{\gamma_\rho}} \, d\tau,$$

where $\Gamma(\cdot)$ is the Gamma function. This expression shows that fractional calculus is nonlocal and has a memory-like effect in dynamical systems. Although this increases the degrees of freedom in the controller design, the influence on stability robustness should be carefully tuned for each specific plant.

In contrast to classic integer-order sliding surfaces, which are usually constructed by the tracking error and its first derivative, the proposed surface includes a fractional-order derivative term. This is advantageous in several aspects: - Tracks past error dynamics, leading to improved stability under constant disturbances. - Enhances error decay and guarantees finite-time convergence in contrast to traditional linear surfaces. - Combining fractional dynamics with the super-twisting algorithm damps the high-frequency oscillations. - The fractional order γ_ρ provides an additional tuning parameter, enabling a trade-off between convergence speed and robustness.

Therefore, the fractional-order super-twisting sliding manifold is a new generalization of classical sliding manifolds, incorporating fractional calculus into nonlinear super-twisting dynamics to achieve better performance in under-actuated quadcopter trajectory tracking.

Remark 1. It is worth noting that the classical super-twisting algorithm (STA) was initially introduced for integer-order systems. In the present work, STA is generalized to fractional order by incorporating Caputo fractional derivatives into the sliding manifold and the reachability law. This extension is theoretically grounded as fractional Lyapunov stability theory proves that inequalities like

$$D^\gamma V(t) \le -aV(t) - bV(t)^\alpha, \quad a, b > 0, \ 0 < \alpha < 1,$$

under Caputo derivatives, guarantee finite-time convergence [46,47]. Besides, the fractional-order extension introduces memory effects, which improve robustness against persistent disturbances and complement the intrinsic robustness of STA. Therefore, the finite-time convergence and robustness properties of STA are preserved in the fractional-order setting, making the proposed fractional-order STA a rigorous and valid extension of the classical algorithm.

The first and second time derivatives of the sliding manifold are:

$$\begin{aligned} \dot{s}_\rho &= \ddot{\varepsilon}_\rho + \alpha_\rho \dot{\varepsilon}_\rho + \beta_\rho D^{\gamma_\rho} \dot{\varepsilon}_\rho, \\ \ddot{s}_\rho &= \dddot{\varepsilon}_\rho + \alpha_\rho \ddot{\varepsilon}_\rho + \beta_\rho D^{\gamma_\rho} \ddot{\varepsilon}_\rho, \end{aligned} \tag{5}$$

where $\dot{\varepsilon}_\rho = \dot{\rho} - \dot{\rho}_d$ and $\ddot{\varepsilon}_\rho = \ddot{\rho} - \ddot{\rho}_d$. The internal dynamics incorporate a positive scalar g and a state-dependent function B_ρ, which serves as a

virtual control input. The control objective is to design B_ρ such that the system dynamics achieve the following structure:

$$\begin{aligned}\dot{\varepsilon}_{1\rho} &= \varepsilon_{2\rho},\\ \dot{\varepsilon}_{2\rho} &= -\dot{\varepsilon}_{1\rho} - \alpha_\rho \varepsilon_{1\rho} - \beta_\rho D^{\gamma_\rho}\varepsilon_{1\rho}.\end{aligned} \tag{6}$$

This structure is achieved by defining a sliding manifold:

$$\sigma_\rho = gB_\rho - D^2\rho_d + s_\rho. \tag{7}$$

Enforcing the sliding mode ensures that $\sigma_\rho \to 0$ in finite time, which leads to the desired dynamics in Eqn. (6). The time derivatives of the sliding variable σ_ρ are:

$$\begin{aligned}D^1\sigma_\rho &= gD^1B_\rho - D^3\rho_d + D^1 s_\rho,\\ D^2\sigma_\rho &= gD^2B_\rho - D^4\rho_d + D^2 s_\rho.\end{aligned} \tag{8}$$

The second derivative of B_ρ is computed as:

$$\begin{aligned}D^2B_\rho &= D^1\left(\frac{\partial B_\rho}{\partial\varphi}D^1\varphi + \frac{\partial B_\rho}{\partial\vartheta}D^1\vartheta + \frac{\partial B_\rho}{\partial\psi}D^1\psi\right)\\ &= \frac{\partial B_\rho}{\partial\varphi}D^2\varphi + \frac{\partial B_\rho}{\partial\vartheta}D^2\vartheta + \frac{\partial B_\rho}{\partial\psi}D^2\psi + D^1\left(\frac{\partial B_\rho}{\partial\vartheta}\right)\dot{\vartheta}\\ &\quad + D^1\left(\frac{\partial B_\rho}{\partial\varphi}\right)\dot{\varphi} + D^1\left(\frac{\partial B_\rho}{\partial\psi}\right)\dot{\psi}.\end{aligned} \tag{9}$$

To enforce the sliding mode, a hierarchical sliding manifold is defined:

$$\bar{\sigma}_\rho = D^1\sigma_\rho + \bar{\alpha}_\rho\sigma_\rho + \bar{\beta}_\rho D^{\bar{\gamma}_\rho}\sigma_\rho, \tag{10}$$

where $\bar{\alpha}_\rho > \bar{\beta}_\rho > 0$ and $0 < \bar{\gamma}_\rho < 1$. The modified super-twisting reachability law is designed as:

$$D^{\bar{\gamma}_\rho}\bar{\sigma}_\rho = -\kappa_{1\rho}\bar{\sigma}_\rho - \kappa_{2\rho}|\bar{\sigma}_\rho|^{1/2}\, sign\,(\bar{\sigma}_\rho) \tag{11}$$

where $\kappa_{1\rho}$ and $\kappa_{2\rho}$ are positive constants and D^{γ_ρ} is the Caputo fractional derivative of order $0 < \gamma_\rho < 1$.

Remark 2. The Caputo definition is adopted because it allows initial conditions to be expressed in terms of integer-order derivatives, which are physically meaningful for dynamical systems, and it is widely used in fractional Lyapunov stability analysis [48]. Unlike the Riemann– Liouville derivative that requires fractional initial states, or the Grünwald–Letnikov derivative mainly suited for numerical approximation, Caputo is more practical for control design and Lyapunov-based stability analysis [46].

Theorem 1. *Consider the $(\rho - \phi)$ under-actuated subsystem dynamics. To enforce sliding mode behavior, a hierarchical fractional-order sliding manifold is introduced as in (10), along with the associated reachability law defined in (11). Based on these formulations, the complete fractional-order control input U_2 that guarantees finite-time convergence is derived as:*

$$\begin{aligned}U_2 = \frac{1}{b_1}\Bigg(&\frac{-1}{g\frac{\partial B_\rho}{\partial\varphi}}\Bigg(g\left(\frac{\partial B_\rho}{\partial\vartheta}D^2\vartheta + \frac{\partial B_\rho}{\partial\psi}D^2\psi + D^1\left(\frac{\partial B_\rho}{\partial\vartheta}\right)\dot{\vartheta}\right.\\ &\left. + D^1\left(\frac{\partial B_\rho}{\partial\varphi}\right)\dot{\varphi} + D^1\left(\frac{\partial B_\rho}{\partial\psi}\right)\dot{\psi}\right) + D^2 s_\rho\\ &+ \bar{\alpha}_\rho D^1\sigma_\rho + \bar{\beta}_\rho D^{\bar{\gamma}_\rho+1}\sigma_\rho + \kappa_{1\rho}\bar{\sigma}_\rho + \kappa_{2\rho}|\bar{\sigma}_\rho|^{1/2}\, sign\,(\bar{\sigma}_\rho)\Bigg)\\ &- (a_1\psi_2\vartheta_2 + a_2\bar{\omega}\vartheta_2)\Bigg).\end{aligned} \tag{12}$$

The designed control input U_2 ensures that the system trajectories reach the sliding manifold in finite time and maintain adherence thereafter, despite the influence of matched uncertainties and external disturbances.

Proof. The finite-time stability of the proposed control law is established through Lyapunov analysis. Consider the Lyapunov candidate function:

$$V_\rho = \frac{1}{2}\bar{\sigma}_\rho^2, \tag{13}$$

where $\bar{\sigma}_\rho$ is the hierarchical sliding manifold defined in (10). Differentiating V_ρ with respect to time in the fractional-order sense gives:

$$D^{\gamma_\rho}V_\rho = \bar{\sigma}_\rho D^{\gamma_\rho}\bar{\sigma}_\rho, \tag{14}$$

where D^{γ_ρ} denotes the Caputo fractional derivative of order $0 < \gamma_\rho < 1$.

Substituting the derivative of the hierarchical sliding manifold (10) along with the control law (12), we obtain:

$$D^{\gamma_\rho}V_\rho = -\kappa_{1\rho}\bar{\sigma}_\rho^2 - \kappa_{2\rho}|\bar{\sigma}_\rho|^{3/2}. \tag{15}$$

Recalling that $V_\rho = \frac{1}{2}\bar{\sigma}_\rho^2$, we can express $|\bar{\sigma}_\rho|$ and $|\bar{\sigma}_\rho|^{3/2}$ in terms of V_ρ, e.g., $|\bar{\sigma}_\rho| = \sqrt{2V_\rho}$, $\quad |\bar{\sigma}_\rho|^{3/2} = (2V_\rho)^{3/4}$. Substituting these into the expression yields:

$$\begin{aligned}&D^{\gamma_\rho}V_\rho = -2\kappa_{1\rho}V_\rho - \kappa_{2\rho}(2V_\rho)^{3/4},\\ &D^{\gamma_\rho}V_\rho + \eta_{1\rho}V_\rho + \eta_{2\rho}V_\rho^{3/4} \le 0,\end{aligned} \tag{16}$$

where $\eta_{1\rho} = 2\kappa_{1\rho}$ and $\eta_{2\rho} = \kappa_{2\rho}(2)^{3/4}$. □

Remark 3. The inequality in (16) is consistent with fractional finite-time stability theory. Specifically, if a Lyapunov function satisfies

$$D^\gamma V(t) \le -aV(t) - bV(t)^\alpha, \quad a, b > 0,\ 0 < \alpha < 1, \tag{17}$$

under the Caputo derivative definition, then $V(t)$ converges to zero in finite time [46,47]. This ensures that V_ρ (and thus $\bar{\sigma}_\rho$) converges to zero in finite time, establishing the stability of the proposed control law.

The fractional-order finite-time convergence bound can be expressed as:

$$T_{f_\rho} \le \frac{1}{\eta_{1\rho}}\ln\left(\frac{\eta_{1\rho}V_\rho^{1/4}(0) + \eta_{2\rho}}{\eta_{2\rho}}\right) \tag{18}$$

This estimate holds under the assumptions that: 1) system states remain bounded, 2) control gains $\kappa_{1\rho}, \kappa_{2\rho}$ are strictly positive, and 3) the fractional order satisfies $0 < \gamma_\rho < 1$.

Therefore, the finite-time convergence of the sliding variable $\bar{\sigma}_\rho$ to zero implies that robust trajectory tracking is ensured in the presence of matched uncertainties and external disturbances.

Starting with a similar procedure, the control law for the $(\hbar - \vartheta)$ under-actuated subsystem, which is governed by Eqn. (3), is obtained. Hence, the expression for U_3 is given by:

$$\begin{aligned}U_3 = \frac{1}{b_2}\Bigg(&\frac{-1}{g\frac{\partial B_\hbar}{\partial\vartheta}}\Bigg(g\left(\frac{\partial B_\hbar}{\partial\varphi}D^2\varphi + \frac{\partial B_\hbar}{\partial\psi}D^2\psi + D^1\left(\frac{\partial B_\hbar}{\partial\vartheta}\right)\dot{\vartheta}\right.\\ &\left. + D^1\left(\frac{\partial B_\hbar}{\partial\varphi}\right)\dot{\varphi} + D^1\left(\frac{\partial B_\hbar}{\partial\psi}\right)\dot{\psi}\right) + D^2 s_\hbar\\ &+ \bar{\alpha}_\hbar D^1\sigma_\hbar + \bar{\beta}_\hbar D^{\bar{\gamma}_\hbar+1}\sigma_\hbar + \kappa_{1\hbar}\bar{\sigma}_\hbar + \kappa_{2\hbar}|\bar{\sigma}_\hbar|^{1/2}\, sign\,(\bar{\sigma}_\hbar)\Bigg)\\ &- (a_3\psi_2\varphi_2 + a_4\bar{\omega}\varphi_2)\Bigg)\end{aligned} \tag{19}$$

A strong basis for reliable quadcopter trajectory tracking is provided by this control law, which guarantees finite-time stabilization of the $(\hbar - \vartheta)$ under-actuated subsystem.

The control design for the fully-actuated subsystem is detailed in the subsection that follows.

3.1.2. Control design of the fully-actuated subsystem

The goal of this subsection is to design the control inputs U_1 and U_4 such that, despite model uncertainties and external disturbances, the tracking errors are driven to zero in finite time. Hence, for the purpose of ensuring finite time convergence, fractional-order super-twisting sliding variables s_z and s_ψ are defined as:

$$\begin{aligned}s_z &= \dot{\varepsilon}_z + \alpha_z\varepsilon_z + \beta_z D^{\gamma_z}\varepsilon_z,\\ s_\psi &= \dot{\varepsilon}_\psi + \alpha_\psi\varepsilon_\psi + \beta_\psi D^{\gamma_\psi}\varepsilon_\psi,\end{aligned} \tag{20}$$

where $\alpha_z, \alpha_\psi > 0$ are linear gains, $\beta_z, \beta_\psi > 0$ are fractional gains, and $0 < \gamma_z, \gamma_\psi < 1$ are fractional orders, the terms D^{γ_z} and D^{γ_ψ} represent the

Caputo fractional derivatives of orders γ_z and γ_ψ, and $\varepsilon_z = z_1 - z_d$ and $\varepsilon_\psi = \psi_1 - \psi_d$ are the tracking errors for vertical position z and yaw angle ψ. The fractional time derivatives of the sliding variables are computed as:

$$\begin{aligned} D^1 s_z &= D^2\varepsilon_z + \alpha_z D^1\varepsilon_z + \beta_z D^{\gamma_z+1}\varepsilon_z, \\ D^1 s_\psi &= D^2\varepsilon_\psi + \alpha_\psi D^1\varepsilon_\psi + \beta_\psi D^{\gamma_\psi+1}\varepsilon_\psi \end{aligned} \tag{21}$$

Substituting the system dynamics from Eqn. (2) into Eqn. (21), we obtain:

$$\begin{aligned} D^1 s_z &= -g + b_0\cos\vartheta_1\cos\varphi_1 U_1 + \Delta_z - D^2 z_d + \alpha_z D^1\varepsilon_z + \beta_z D^{\gamma_z+1}\varepsilon_z, \\ D^1 s_\psi &= a_5\vartheta_2\varphi_2 + b_3 U_4 + \Delta_\psi - D^2\psi_d + \alpha_\psi D^1\varepsilon_\psi + \beta_\psi D^{\gamma_\psi+1}\varepsilon_\psi \end{aligned} \tag{22}$$

where Δ_z and Δ_ψ represent lumped uncertainties and disturbances affecting the z and ψ-dynamics, respectively.

To enforce the sliding mode, the following fractional reachability laws are designed:

$$\begin{aligned} D^1 s_z &= -\kappa_{1z}\bar{s}_z - \kappa_{2z}|\bar{s}_z|^{1/2}\, sign\,(\bar{s}_z), \\ D^1 s_\psi &= -\kappa_{1\psi}\bar{s}_\psi - \kappa_{2\psi}|\bar{s}_\psi|^{1/2}\, sign\,(\bar{s}_\psi) \end{aligned} \tag{23}$$

where $\kappa_{1z}, \kappa_{2z}, \kappa_{1\psi}, \kappa_{2\psi} > 0$ are positive constants. These laws ensure that the sliding variables s_z and s_ψ converge to zero in finite time, thereby enforcing the sliding mode.

Theorem 2. *Consider the fully actuated subsystem (2) with the fractional derivatives of the sliding variables defined in (22) and governed by the reachability laws (23). The control inputs U_1 and U_4 are derived as follows:*

$$\begin{aligned} U_1 &= \frac{1}{b_0\cos\vartheta_1\cos\varphi_1}\big(g - \kappa_{1z}\bar{s}_z - \kappa_{2z}|\bar{s}_z|^{1/2}\, sign\,(\bar{s}_z) + D^2 z_d - \alpha_z D^1\varepsilon_z \\ &\quad - \beta_z D^{\gamma_z+1}\varepsilon_z\big), \\ U_4 &= \frac{1}{b_3}\big(-a_5\vartheta_2\varphi_2 - \kappa_{1\psi}\bar{s}_\psi - \kappa_{2\psi}|\bar{s}_\psi|^{1/2}\, sign\,(\bar{s}_\psi) + D^2\psi_d - \alpha_\psi D^1\varepsilon_\psi \\ &\quad - \beta_\psi D^{\gamma_\psi+1}\varepsilon_\psi\big) \end{aligned} \tag{24}$$

These control inputs ensure that the sliding variables s_z and s_ψ converge to zero in finite time, thereby achieving finite-time tracking of the desired trajectories for the quadcopter system.

Proof. To prove the finite-time convergence of the sliding variables and tracking errors, consider the Lyapunov functions $V_z = \frac{1}{2}s_z^2$ and $V_\psi = \frac{1}{2}s_\psi^2$. Their time derivatives along the system dynamics are:

$$\begin{aligned} D^1 V_z &= \bar{s}_z\big(-g + b_0\cos\vartheta_1\cos\varphi_1 U_1 + \Delta_z - D^2 z_d + \alpha_z D^1\varepsilon_z + \beta_z D^{\gamma_z+1}\varepsilon_z\big), \\ D^1 V_\psi &= \bar{s}_\psi\big(a_5\vartheta_2\varphi_2 + b_3 U_4 + \Delta_\psi - D^2\psi_d + \alpha_\psi D^1\varepsilon_\psi + \beta_\psi D^{\gamma_\psi+1}\varepsilon_\psi\big) \end{aligned} \tag{25}$$

Substituting the control inputs from Eqn. (24) and rearranging, we obtain:

$$\begin{aligned} D^1 V_z &\le -\kappa_{1z}\bar{s}_z^2 - \lambda_z|\bar{s}_z|^{3/2}, \\ D^1 V\psi &\le -\kappa_{1\psi}\bar{s}_\psi^2 - \lambda_\psi|\bar{s}_\psi|^{3/2} \end{aligned} \tag{26}$$

where $\lambda_z = \kappa_{2z} - |\Delta_z|$ and $\lambda_\psi = \kappa_{2\psi} - |\Delta_\psi|$. These inequalities ensure finite-time convergence of the sliding variables to zero. The differential inequalities in Eqn. (26) can be rewritten as:

$$\begin{aligned} D^1 V_z + \eta_{1z}V_z + \eta_{2z}V_z^{3/4} &\le 0, \\ D^1 V_\psi + \eta_{1\psi}V_\psi + \eta_{2\psi}V_\psi^{3/4} &\le 0 \end{aligned} \tag{27}$$

where $\eta_{1z} = 2\kappa_{1z}$, $\eta_{2z} = \sqrt{2}\lambda_z$, $\eta_{1\psi} = 2\kappa_{1\psi}$, and $\eta_{2\psi} = \sqrt{2}\lambda_\psi$. The settling times for s_z and s_ψ are given by:

$$\begin{aligned} T_{f_z} &\le \frac{1}{\eta_{1z}}\ln\left(\frac{\eta_{1z}V_z^{1/4}(0) + \eta_{2z}}{\eta_{2z}}\right), \\ T_{f_\psi} &\le \frac{1}{\eta_{1\psi}}\ln\left(\frac{\eta_{1\psi}V_\psi^{1/4}(0) + \eta_{2\psi}}{\eta_{2\psi}}\right) \end{aligned} \tag{28}$$

These expressions confirm that the sliding variables s_z and s_ψ converge to zero in finite time, ensuring finite-time tracking of the desired trajectories. Once the sliding mode is enforced, the tracking errors ε_z and ε_ψ satisfy the following dynamics:

$$\begin{aligned} D^2\varepsilon_z + \alpha_z D^1\varepsilon_z + \beta_z D^{\gamma_z+1}\varepsilon_z &= 0, \\ D^2\varepsilon_\psi + \alpha_\psi D^1\varepsilon_\psi + \beta_\psi D^{\gamma_\psi+1}\varepsilon_\psi &= 0 \end{aligned} \tag{29}$$

These equations are finite-time stable, ensuring that ε_z and ε_ψ converge to zero in finite time. □

Remark 4. High-precision differentiators and the integration of neural networks increase the performance of the controller, especially for real-world applications. In practical realization, nonlinear drift terms F and H should be estimated by a neural network. For being robust against model uncertainties and external disturbances, these terms have to be replaced by their estimates $\hat{F}$ and $\hat{H}$ both in simulations and real-world applications.

3.2. Neuro-adaptive fractional order super- twisting sliding mode control design

This subsection presents the neuro- adaptive fractional- order super-twisting sliding mode control framework, which integrates a deep residual neural network with attention mechanisms to enhance precision and adaptability. The functions (H_z, H_ψ) and (F_z, F_ψ), which represent the unknown nonlinear drift and distribution, are approximated via the network, making it challenging to represent by means of simple feed-forward networks because of the high complexity and nonlinearities of the dynamics of quadcopter. Through the use of residual learning for efficient feature extraction and attention mechanisms to attend to important system dynamics, the proposed method achieves remarkable improvements in approximation accuracy, robustness, and flexibility in the presence of uncertainties. The NA-FOSTSMC framework also alleviates the vanishing gradient problem, ensuring robust performance in real-world applications. Its adaptive learning capability allows attention weights to evolve over time in response to changing system dynamics, making it a suitable solution for quadcopter regulation under dynamic and unpredictable conditions.

3.2.1. Neural network architecture

The DRNAM introduces several novelties in the context of traditional neural network architecture. The DRNAM architecture (in Figure 2) consists of four residual blocks and two fully connected layers of 64 neurons each, with ReLU as the activation function. To facilitate residual learning, skip connections are used, which enhance training stability and alleviate the vanishing gradient problem. A self-attention layer is applied after each residual block to adaptively focus on important input features based on system variations. This structure enables the network to capture high-order nonlinearities while remaining computationally efficient.

Fig. 3 depicts the tracking of the quadcopter's reference trajectory under the suggested control approach. It can be seen from the results that the tracking error is considerably small and the convergence is much faster using both with and without FOSTSMD and DRNAM. This is further verified from Figs. 4–6, wherein the translational states (x, y, z) reach and maintain the reference trajectory in finite time.

Input Layer: The input vector $\mathbf{X} \in \mathbb{R}^n$ consists of the quadcopter's position and orientation states and their derivatives:

$$\mathbf{X} = [x_1, x_2, y_1, y_2, z_1, z_2, \varphi_1, \varphi_2, \vartheta_1, \vartheta_2, \psi_1, \psi_2]^T. \tag{30}$$

Residual Blocks: Each residual block computes:

$$\mathbf{R}_k = \mathcal{F}_k(\mathbf{R}_{k-1}) + \mathbf{R}_{k-1}, \quad k = 1, 2, \ldots, 4 \tag{31}$$

where $\mathcal{F}_k$ is the nonlinear transformation learned by the k-th block. The residual architecture leads to better generalization and less overfitting

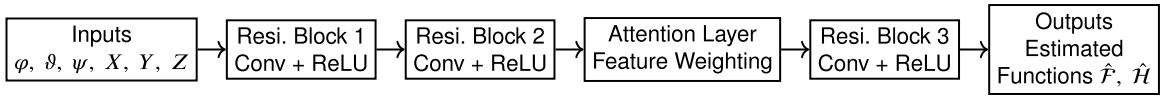

Fig. 2. DRNAM Network module to estimate the unfamiliar nonlinear distortions from quadcopter states..

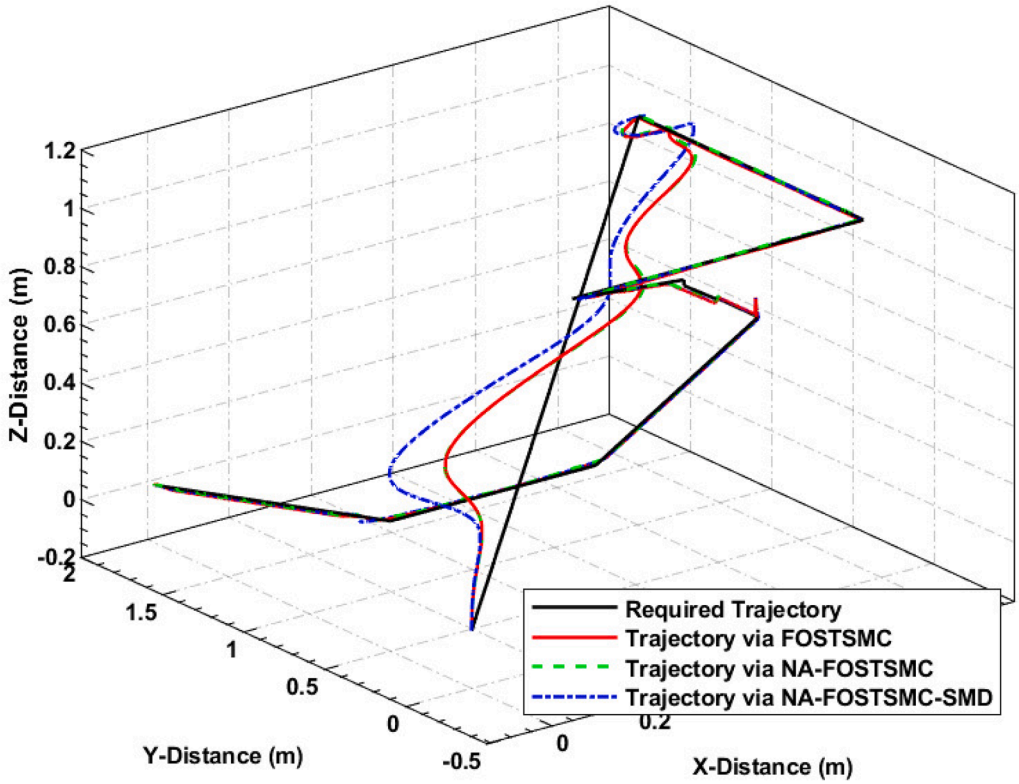

Fig. 3. 3D flight path tracking performance of the quadcopter under the neuro-adaptive FOSTSMC controller, demonstrating high-precision trajectory following in spatial coordinates.

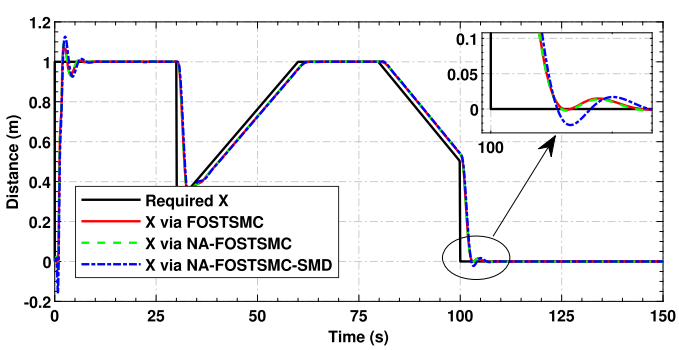

Fig. 4. x-axis trajectory tracking performance using the NA-FOSTSMC scheme enhanced with DRNAM and SMD, demonstrating precise and robust control behaviour.

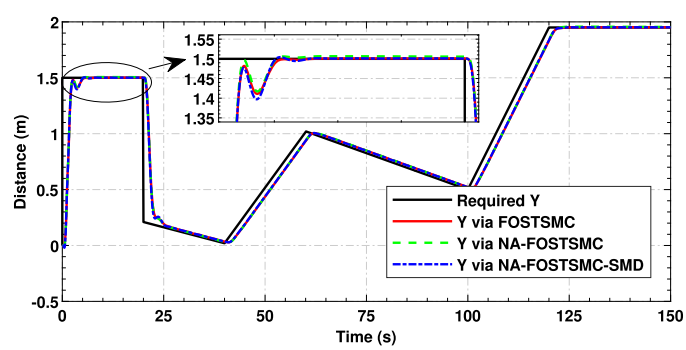

Fig. 5. Trajectory tracking performance of the proposed NA-FOSTSMC scheme for the y-axis, incorporating DRNAM and SMD mechanisms.

than standard feedforward networks.

Attention Mechanism: Attention weights α_i are computed using:

$$\mathbf{A}_\vartheta = \sum_{i=1}^{n} \alpha_i \mathbf{R}_k^{(i)}, \quad \alpha_i = \frac{\exp(\mathbf{v}^T \mathbf{R}_K^{(i)})}{\sum_{j=1}^{n} \exp(\mathbf{v}^T \mathbf{R}_K^{(j)})}, \tag{32}$$

where $\mathbf{v}$ is a learnable attention vector. This mechanism enables the network to dynamically focus on the important features of the states (e.g., orientation in the presence of disturbances), which is not feasible in traditional residual networks, and it quantitatively enhances the performance as this yields: smaller approximation error, faster convergence, and stronger robustness when compared with a normal residual NN.

Output Layer: The final outputs are:

$$\hat{F}_z = \mathbf{W}_F^T \mathbf{A}_\vartheta + b_F, \quad \hat{H}_z = \mathbf{W}_H^T \mathbf{A}_\vartheta + b_H, \tag{33}$$

where $\mathbf{W}_F, \mathbf{W}_H \in \mathbb{R}^{64}$ and b_F, b_H are biases. The resulting outputs represent the estimated drift and distribution functions, enabling adaptive compensation of the nonlinear quadcopter dynamics.

3.2.2. Training and adaptation

The network is optimized by Adam with a learning rate of 1×10^{-3}. The training set is produced from simulated quadcopter trajectories with

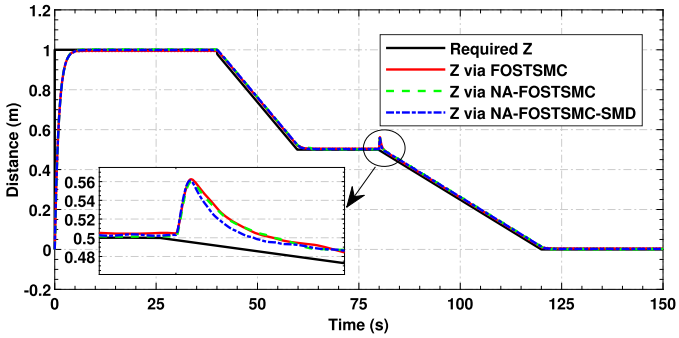

Fig. 6. Trajectory tracking performance of the proposed NA-FOSTSMC scheme for the z-axis, incorporating DRNAM and SMD mechanisms.

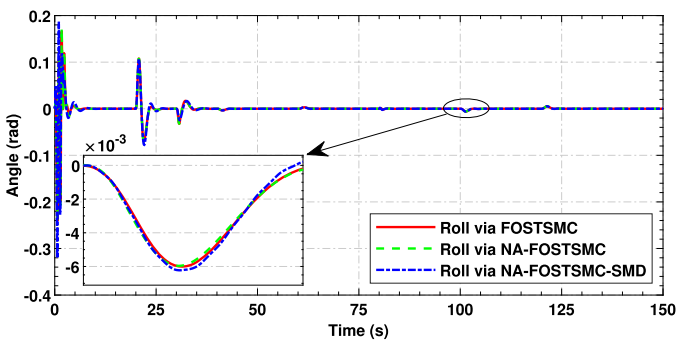

Fig. 7. Roll subsystem response using the NA-FOSTSMC scheme, incorporating DRNAM and SMD for enhanced tracking performance.

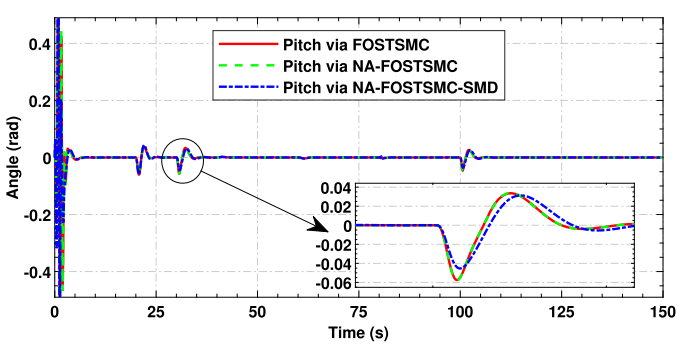

Fig. 8. Pitch subsystem dynamics under the NA-FOSTSMC scheme utilizing DRNAM and SMD for improved control precision.

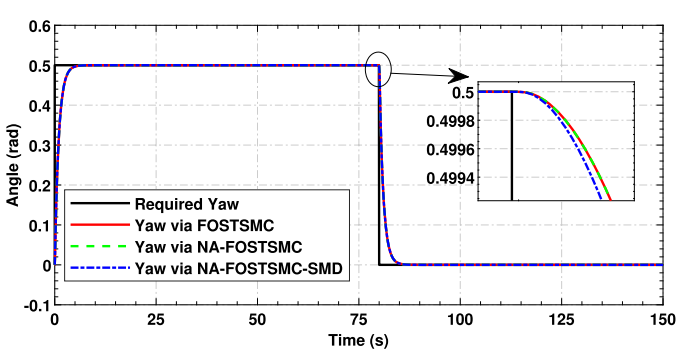

Fig. 9. Response of the yaw subsystem controlled by the NA-FOSTSMC approach with DRNAM and SMD enhancements.

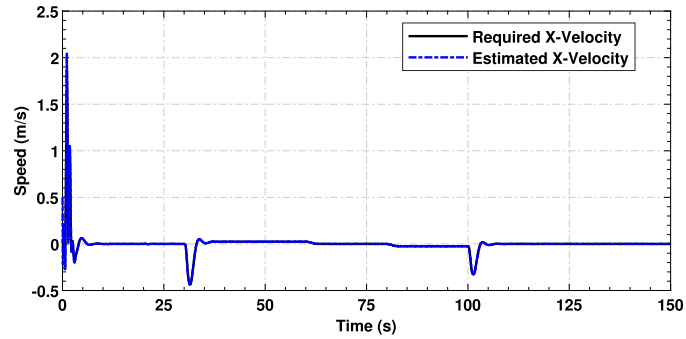

Fig. 10. x-velocity estimation with SMD.

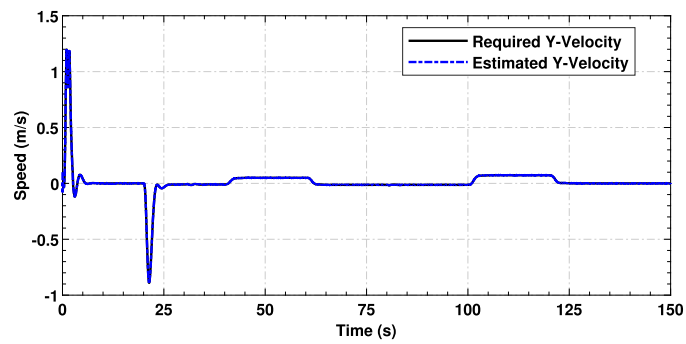

Fig. 11. y-velocity estimation with SMD.

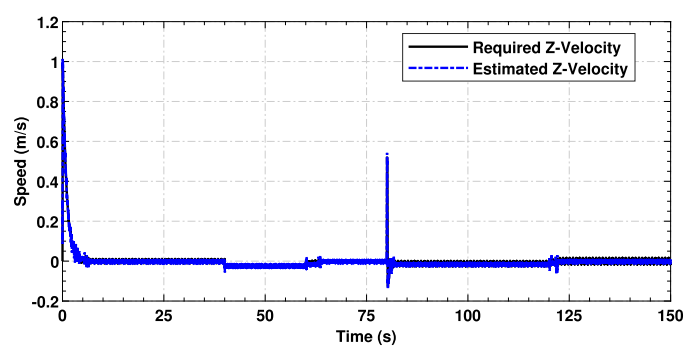

Fig. 12. z-velocity estimation with SMD.

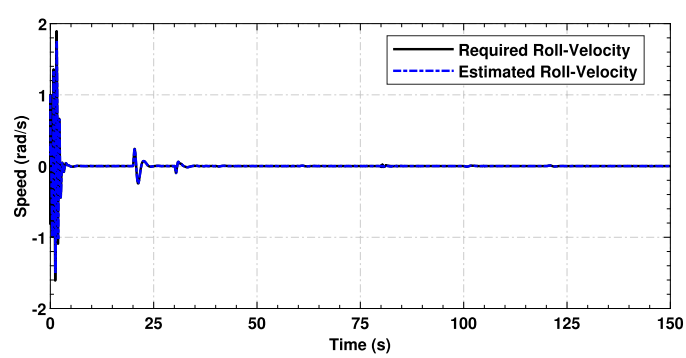

Fig. 13. Roll velocity estimation with SMD.

different disturbances and non-linear dynamics. Training is performed for 200 epochs, with early stopping if the validation loss does not improve for 20 consecutive epochs. Loss function is given by:

$$\mathcal{L} = \frac{1}{2}\left(\|\hat{F}_z - F_z\|^2 + \|\hat{H}_z - H_z\|^2\right). \tag{34}$$

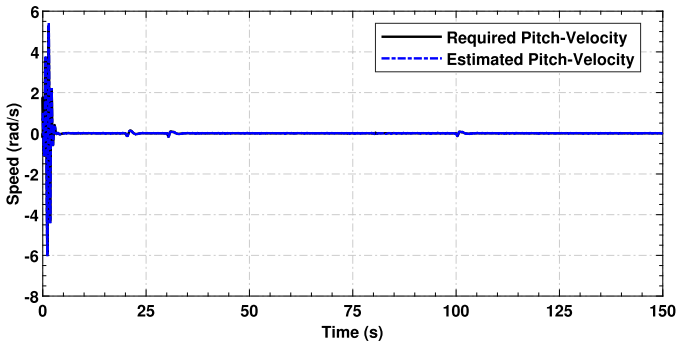

Fig. 14. Pitch velocity estimation with SMD.

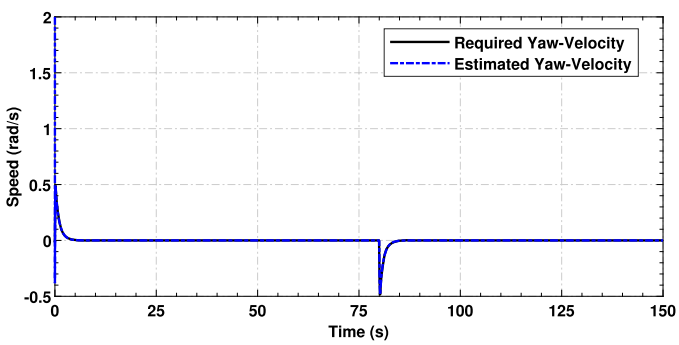

Fig. 15. Yaw velocity estimation with SMD.

During training, the weights are updated with backpropagation, and the attention mechanism modulates the importance of features as a function of the approximation error. After being trained, the network is able to adapt to new input data by adjusting the attention weights, thus it can provide an accurate approximation even in the presence of time varying system dynamics.

Remark 5. The novelty of the proposed DRNAM lies in combining residual learning and self-attention for adaptive feature prioritization. Unlike traditional neural networks or regular residual architectures, the DRNAM can adaptively assign greater importance to the most critical state variables under disturbance, and its residual connections help stabilize the training process. The combined novelty enables a strong approximation of the nonlinear quadcopter dynamics with reduced reliance on accurate modeling, thereby improving robustness and control performance.

3.2.3. Stability analysis

To guarantee the stability and robustness of the control system with the DRNAM, a comprehensive stability analysis is conducted using Lyapunov theory. The analysis also accounts for the approximation errors introduced by the neural network.

The system dynamics for each state variable $\mathfrak{z}$ can be represented by the following expression.

$$\begin{aligned}\dot{\mathfrak{z}}_1 &= \mathfrak{z}_2,\\ \dot{\mathfrak{z}}_2 &= F_{\mathfrak{z}}(t,\mathfrak{z}) + H_{\mathfrak{z}}(t,\mathfrak{z})U_{\mathfrak{z}}(t,\mathfrak{z}),\end{aligned} \tag{35}$$

where $\mathfrak{z}_1$ is the state variable (e.g., $x, y, z, \varphi, \vartheta, \psi$), $\mathfrak{z}_2$ is the derivative of the state variable (e.g., $\dot{x}, \dot{y}, \dot{z}, \dot{\varphi}, \dot{\vartheta}, \dot{\psi}$), $F_{\mathfrak{z}}$ represents the system dynamics, $H_{\mathfrak{z}}$ represents the input dynamics and $U_{\mathfrak{z}}$ is the control input. The sliding variable σ is defined as:

$$\sigma_{\mathfrak{z}} = \dot{\varepsilon}_{\mathfrak{z}} + \alpha_{\mathfrak{z}}\varepsilon_{\mathfrak{z}} + \beta_{\mathfrak{z}}D^{\gamma_{\mathfrak{z}}}\varepsilon_{\mathfrak{z}}, \tag{36}$$

where $\varepsilon_{\mathfrak{z}} = \mathfrak{z}_1 - \mathfrak{z}_d$ is the tracking error, and $\alpha_{\mathfrak{z}}, \beta_{\mathfrak{z}} > 0$ are positive design constants. The operator $D^{\gamma_{\mathfrak{z}}}$ represents the fractional derivative of order $\gamma_{\mathfrak{z}}$, where $0 < \gamma_{\mathfrak{z}} < 1$. The time derivative of the sliding variable is expressed as:

$$\begin{aligned}\dot{\sigma}_{\mathfrak{z}} &= \ddot{\varepsilon}_{\mathfrak{z}} + \alpha_{\mathfrak{z}}\dot{\varepsilon}_{\mathfrak{z}} + \beta_{\mathfrak{z}}D^{\gamma_{\mathfrak{z}}}\dot{\varepsilon}_{\mathfrak{z}}.\\ \dot{\sigma}_{\mathfrak{z}} &= \dot{\mathfrak{z}}_2 - \ddot{\mathfrak{z}}_d + \alpha_{\mathfrak{z}}\dot{\varepsilon}_{\mathfrak{z}} + \beta_{\mathfrak{z}}D^{\gamma_{\mathfrak{z}}}\dot{\varepsilon}_{\mathfrak{z}}\\ \dot{\sigma}_{\mathfrak{z}} &= F_{\mathfrak{z}}(t,\mathfrak{z}) + H_{\mathfrak{z}}(t,\mathfrak{z})U_{\mathfrak{z}}(t,\mathfrak{z}) - \ddot{\mathfrak{z}}_d + \alpha_{\mathfrak{z}}\dot{\varepsilon}_{\mathfrak{z}} + \beta_{\mathfrak{z}}D^{\gamma_{\mathfrak{z}}}\dot{\varepsilon}_{\mathfrak{z}}\end{aligned} \tag{37}$$

Theorem 3. *Consider the system dynamics (35) with unknown nonlinear drift and distribution functions $F_{\mathfrak{z}}(t,\mathfrak{z})$ and $H_{\mathfrak{z}}(t,\mathfrak{z})$, approximated by a DRNAM. Define the control law as:*

$$U_{\mathfrak{z}} = \frac{1}{\hat{H}_{\mathfrak{z}}(t,\mathfrak{z})}\left(\ddot{\mathfrak{z}}_d - \hat{F}_{\mathfrak{z}}(t,\mathfrak{z}) - \alpha_{\mathfrak{z}}\dot{\varepsilon}_{\mathfrak{z}} - \beta_{\mathfrak{z}}D^{\gamma_{\mathfrak{z}}}\dot{\varepsilon}_{\mathfrak{z}} - \kappa_1\sigma_{\mathfrak{z}} - \kappa_2\, sign\,(\sigma_{\mathfrak{z}})\right), \tag{38}$$

where $\kappa_1, \kappa_2 > 0$ are positive design constants. The neural network approximations are given by:

$$\hat{F}_{\mathfrak{z}}(t,\mathfrak{z}) = \mathbf{W}_{F_{\mathfrak{z}}}^T\mathbf{A}_{\mathfrak{z}} + b_{F_{\mathfrak{z}}}, \quad \hat{H}_{\mathfrak{z}}(t,\mathfrak{z}) = \mathbf{W}_{H_{\mathfrak{z}}}^T\mathbf{A}_{\mathfrak{z}} + b_{H_{\mathfrak{z}}}, \tag{39}$$

where $\mathbf{A}_{\mathfrak{z}}$ denotes the attention-weighted output of the residual blocks. Let the approximation errors be:

$$\tilde{F}_{\mathfrak{z}} = \hat{F}_{\mathfrak{z}} - F_{\mathfrak{z}}, \quad \tilde{H}_{\mathfrak{z}} = \hat{H}_{\mathfrak{z}} - H_{\mathfrak{z}}, \tag{40}$$

with $|\tilde{F}_{\mathfrak{z}}| \le \bar{F}_{\mathfrak{z}}$ and $|\tilde{H}_{\mathfrak{z}}| \le \bar{H}_{\mathfrak{z}}$ for some positive constants $\bar{F}_{\mathfrak{z}}$ and $\bar{H}_{\mathfrak{z}}$. Then, under the proposed control law (38), the sliding variable $\sigma_{\mathfrak{z}}$ converges to zero in finite time, ensuring finite-time stability of the closed-loop system.

Proof. To analyze the stability of the system, a Lyapunov function $V_{\mathfrak{z}}$ is defined as:

$$V_{\mathfrak{z}} = \frac{1}{2}\sigma_{\mathfrak{z}}^2 + \frac{1}{2}\tilde{\mathbf{W}}_{F_{\mathfrak{z}}}^T\tilde{\mathbf{W}}_{F_{\mathfrak{z}}} + \frac{1}{2}\tilde{\mathbf{W}}_{H_{\mathfrak{z}}}^T\tilde{\mathbf{W}}_{H_{\mathfrak{z}}}, \tag{41}$$

where $\tilde{\mathbf{W}}_{F_{\mathfrak{z}}} = \hat{\mathbf{W}}_{F_{\mathfrak{z}}} - \mathbf{W}_{F_{\mathfrak{z}}}$ and $\tilde{\mathbf{W}}_{H_{\mathfrak{z}}} = \hat{\mathbf{W}}_{H_{\mathfrak{z}}} - \mathbf{W}_{H_{\mathfrak{z}}}$ are the weight estimation errors, and $\mathbf{W}_{F_{\mathfrak{z}}}$ and $\mathbf{W}_{H_{\mathfrak{z}}}$ are the true weights of the neural network. The time derivative of the Lyapunov function is computed as:

$$\begin{aligned}\dot{V}_{\mathfrak{z}} &= \sigma_{\mathfrak{z}}\dot{\sigma}_{\mathfrak{z}} + \tilde{\mathbf{W}}_{F_{\mathfrak{z}}}^T\dot{\tilde{\mathbf{W}}}_{F_{\mathfrak{z}}} + \tilde{\mathbf{W}}_{H_{\mathfrak{z}}}^T\dot{\tilde{\mathbf{W}}}_{H_{\mathfrak{z}}}\\ \dot{V}_{\mathfrak{z}} &= \sigma_{\mathfrak{z}}\left(F_{\mathfrak{z}} + H_{\mathfrak{z}}U_{\mathfrak{z}} - \ddot{\mathfrak{z}}_d + \alpha_{\mathfrak{z}}\dot{\varepsilon}_{\mathfrak{z}} + \beta_{\mathfrak{z}}D^{\gamma_{\mathfrak{z}}}\dot{\varepsilon}_{\mathfrak{z}}\right) + \tilde{\mathbf{W}}_{F_{\mathfrak{z}}}^T\dot{\tilde{\mathbf{W}}}_{F_{\mathfrak{z}}} + \tilde{\mathbf{W}}_{H_{\mathfrak{z}}}^T\dot{\tilde{\mathbf{W}}}_{H_{\mathfrak{z}}}.\end{aligned} \tag{42}$$

Using the control law (38) and the approximation errors (40), this simplifies to:

$$\dot{V}_{\mathfrak{z}} = \sigma_{\mathfrak{z}}\left(-\kappa_1\sigma - \kappa_2\, sign\,(\sigma_{\mathfrak{z}}) + \tilde{F}_{\mathfrak{z}} + \tilde{H}_{\mathfrak{z}}U_{\mathfrak{z}} + \Delta_{\mathfrak{z}}\right) + \tilde{\mathbf{W}}_{F_{\mathfrak{z}}}^T\dot{\tilde{\mathbf{W}}}_{F_{\mathfrak{z}}} + \tilde{\mathbf{W}}_{H_{\mathfrak{z}}}^T\dot{\tilde{\mathbf{W}}}_{H_{\mathfrak{z}}}. \tag{43}$$

Assuming the disturbances $\Delta_{\mathfrak{z}}$ are also bounded by $|\Delta_{\mathfrak{z}}| \le \bar{\Delta}_{\mathfrak{z}}$, we can bound the term $\sigma_{\mathfrak{z}}(\tilde{F}_{\mathfrak{z}} + \tilde{H}_{\mathfrak{z}}U_{\mathfrak{z}} + \Delta_{\mathfrak{z}})$ as:

$$\sigma_{\mathfrak{z}}(\tilde{F}_{\mathfrak{z}} + \tilde{H}_{\mathfrak{z}}U_{\mathfrak{z}} + \Delta_{\mathfrak{z}}) \le |\sigma_{\mathfrak{z}}|(\bar{F}_{\mathfrak{z}} + \bar{H}_{\mathfrak{z}}|U_{\mathfrak{z}}| + \bar{\Delta}_{\mathfrak{z}}). \tag{44}$$

The weight adaptation laws for the neural network are designed as:

$$\dot{\tilde{\mathbf{W}}}_{F_{\mathfrak{z}}} = -\eta_F\sigma_{\mathfrak{z}}\mathbf{A}_{\mathfrak{z}}, \quad \dot{\tilde{\mathbf{W}}}_{H_{\mathfrak{z}}} = -\eta_H\sigma_{\mathfrak{z}}U_{\mathfrak{z}}\mathbf{A}_{\mathfrak{z}}, \tag{45}$$

where $\eta_F, \eta_H > 0$ are positive learning rates. Substituting these into the Lyapunov derivative, we obtain:

$$\begin{aligned}\dot{V}_{\mathfrak{z}} \le{}& -\kappa_1\sigma_{\mathfrak{z}}^2 - \kappa_2|\sigma_{\mathfrak{z}}| + |\sigma_{\mathfrak{z}}|(\bar{F}_{\mathfrak{z}} + \bar{H}_{\mathfrak{z}}|U_{\mathfrak{z}}| + \bar{\Delta}_{\mathfrak{z}}) - \eta_F\tilde{\mathbf{W}}_{F_{\mathfrak{z}}}^T\sigma_{\mathfrak{z}}\mathbf{A}_{\mathfrak{z}}\\ &- \eta_H\tilde{\mathbf{W}}_{H_{\mathfrak{z}}}^T\sigma U_{\mathfrak{z}}\mathbf{A}_{\mathfrak{z}}.\end{aligned} \tag{46}$$

By choosing the control gains κ_1 and κ_2 such that:

$$\kappa_1 > 0, \quad \kappa_2 > \bar{F}_{\mathfrak{z}} + \bar{H}_{\mathfrak{z}}|U_{\mathfrak{z}}| + \bar{\Delta}_{\mathfrak{z}}, \tag{47}$$

we can ensure that:

$$\dot{V}_{\mathfrak{z}} \le -\kappa_1\sigma_{\mathfrak{z}}^2 - \lambda_{\mathfrak{z}}|\sigma_{\mathfrak{z}}|, \tag{48}$$

where $\lambda_{\mathfrak{z}} = \kappa_2 - (\bar{F}_{\mathfrak{z}} + \bar{H}_{\mathfrak{z}}|U_{\mathfrak{z}}| + \bar{\Delta}_{\mathfrak{z}}) > 0$. Expressing in terms of $V_{\mathfrak{z}}$ ($\sigma_{\mathfrak{z}}^2 = 2V_{\mathfrak{z}}$, $|\sigma_{\mathfrak{z}}| = \sqrt{2V_{\mathfrak{z}}}$):

$$\dot{V}_{\mathfrak{z}} \le -\kappa_1(2V_{\mathfrak{z}}) - \lambda_{\mathfrak{z}}\sqrt{2V_{\mathfrak{z}}} \tag{49}$$

Letting $\eta_1 = 2\kappa_1$ and $\eta_2 = \lambda_\mathfrak{z}\sqrt{2}$:

$$\dot{V}_\mathfrak{z} \leq -\eta_1 V_\mathfrak{z} - \eta_2 V_\mathfrak{z}^{1/2} \tag{50}$$

This inequality ensures that $\sigma_\mathfrak{z}$ converges to zero in finite time, hence guaranteeing the stability of the system. Lyapunov analysis confirms that the DRNAM-based control system enjoys finite-time stability, which ensures $\varepsilon_\mathfrak{z}$ also converges to zero. Moreover, the approximation errors in the neural network remain bounded, and adaptation of weights is done to be accurate for practical implementation in autonomous navigation of quadcopters. □

3.3. Fractional-order super-Twisting sliding mode differentiator design

In this sub-section, herein, the FOST-SMD integrates the STA with the fractional-order calculus to provide robust and precise state variable derivative estimations of $x, y, z, \varphi, \vartheta, \psi$ in the presence of noise and disturbances. The FOST-SMD is also bestowed with some substantial advantages that make it a robust derivative estimation tool for dynamic systems. First, the FOST-SMD is resistant to noise and disturbances in the input signal and is guaranteed to deliver sound performance irrespective of uncertainty. Secondly, the differentiator guarantees finite-time convergence-that is, it provides true derivative estimation in a finite period of time-which is a significant concern for real-time control purposes. Thirdly, fractional-order design allows it to be more flexible, and hence, the differentiator can approximate and process complex systems better than with conventional integer-order methods. Fourth, it is numerically stable and has great ability to prevent numerical chattering, further ensuring the accuracy and reliability of the differentiator.

The differentiator dynamics are given by [49]:

$$\begin{aligned} \dot{\hat{\mathfrak{z}}}_1 &= \xi_1\sqrt{|\mathfrak{z}_1 - \hat{\mathfrak{z}}_1|}\cdot\ sign\,(\mathfrak{z}_1 - \hat{\mathfrak{z}}_1) + \hat{\mathfrak{z}}_2, \\ D^\alpha\hat{\mathfrak{z}}_2 &= \xi_2\cdot\ sign\,(\mathfrak{z}_1 - \hat{\mathfrak{z}}_1), \end{aligned} \tag{51}$$

where $\hat{\mathfrak{z}}_1$ is the estimate of $\mathfrak{z}_1$, $\hat{\mathfrak{z}}_2$ is the estimate of the fractional-order derivative $D^\alpha\mathfrak{z}_1$, ξ_1 and ξ_2 are positive design gains, and D^α is the Riemann-Liouville fractional derivative operator of order α $(0 < \alpha < 1)$. The fractional-order derivative $D^{\gamma_\mathfrak{z}}x(t)$ is defined using the Caputo fractional derivative:

$$D^{\gamma_\mathfrak{z}}x(t) = \frac{1}{\Gamma(1-\gamma)}\int_0^t \frac{\dot{x}(\tau)}{(t-\tau)^{\gamma_\mathfrak{z}}}d\tau \tag{52}$$

where $\Gamma(\cdot)$ is the Gamma function. The ST-FOD approximates this fractional-order derivative using the Super-Twisting Algorithm.

The estimation errors are defined as:

$$\begin{aligned} e_1 &= \mathfrak{z}_1 - \hat{\mathfrak{z}}_1, \\ e_2 &= \mathfrak{z}_2 - \hat{\mathfrak{z}}_2. \end{aligned} \tag{53}$$

The error dynamics are derived as:

$$\begin{aligned} \dot{e}_1 &= -\xi_1\sqrt{|e_1|}\cdot\ sign\,(e_1) + e_2, \\ D^\alpha e_2 &= -\xi_2\cdot\ sign\,(e_1) + D^{1+\alpha}\mathfrak{z}_1. \end{aligned} \tag{54}$$

where $D^\alpha e_2 = D^\alpha\mathfrak{z}_2 - D^\alpha\hat{\mathfrak{z}}_2 = D^{1+\alpha}\mathfrak{z}_1 - D^\alpha\hat{\mathfrak{z}}_2$, $D^{1+\alpha}\mathfrak{z}_1$ represents the fractional-order derivative of $\mathfrak{z}_2$. To simplify the analysis, the error e_2 is decomposed into two components:

$$e_2 = v_i + \phi_i \tag{55}$$

where v_i is a known intermediate variable, and ϕ_i represents the unknown disturbance, with $D^\alpha\phi_i = \Delta_i(t) = D^{1+\alpha}\mathfrak{z}_1$. The error dynamics are rewritten as:

$$\begin{aligned} \dot{e}_1 &= -\xi_1\sqrt{|e_1|}\cdot\ sign\,(e_1) + v_i + \phi_i, \\ D^\alpha e_2 &= D^\alpha v_i + D^\alpha\phi_i, \\ D^\alpha v_i &= -\xi_2\cdot\ sign\,(e_1), \\ D^\alpha\phi_i &= \Delta_i(t). \end{aligned} \tag{56}$$

The FOST-SMD provides a robust and accurate method for estimating the derivatives of state variables $x, y, z, \varphi, \vartheta, \psi$.

3.3.1. Lyapunov stability analysis

To prove the stability of FOST-SMD, we apply Lyapunov theory that aims to prove the convergence of the estimation errors to zero in finite time, hence proving the stability and robustness of the differentiator.

For the stability analysis, we define a Lyapunov function candidate V as:

$$V = \frac{1}{2}e_1^2 + \frac{1}{2}e_2^2 \tag{57}$$

This is the Lyapunov function of the total energy of the estimation errors. The purpose would be to prove that V decreases since time elapses, meaning the convergence of the errors to zero.

The time derivative of the Lyapunov function V is:

$$\dot{V}_\mathfrak{z} = e_1\dot{e}_1 + e_2\dot{e}_2 \tag{58}$$

Substituting the error dynamics $\dot{e}_1$ and $D^\alpha e_2$, we get:

$$\dot{V}_\mathfrak{z} = e_1\Big(-\xi_1\sqrt{|e_1|}\cdot\ sign\,(e_1) + e_2\Big) + e_2\Big(-\xi_2\cdot\ sign\,(e_1) + D^{1+\alpha}\mathfrak{z}_1\Big) \tag{59}$$

The term $e_1\cdot\ sign\,(e_1) = |e_1|$, so the expression simplifies to:

$$\dot{V}_\mathfrak{z} = -\xi_1|e_1|^{3/2} + e_1e_2 - \xi_2|e_2| + e_2D^{1+\alpha}\mathfrak{z}_1 \tag{60}$$

To ensure stability, we need to show that $\dot{V}_\mathfrak{z}$ is negative definite. This requires bounding the terms involving e_2.

The term $e_2D^{1+\alpha}\mathfrak{z}_1$ represents the effect of the fractional-order derivative of the system dynamics. Assuming that $D^{1+\alpha}\mathfrak{z}_1$ is bounded by a constant Δ, we have:

$$|e_2D^{1+\alpha}\mathfrak{z}_1| \leq |e_2|\cdot\Delta. \tag{61}$$

Thus, the Lyapunov derivative can be bounded as:

$$\dot{V}_\mathfrak{z} \leq -\xi_1|e_1|^{3/2} + |e_1|\cdot|e_2| - \xi_2|e_2| + \Delta|e_2| \tag{62}$$

To ensure $\dot{V}_\mathfrak{z}$ is negative definite, the gains ξ_1 and ξ_2 must be chosen such that:

$$\xi_1 > 0, \quad \xi_2 > \Delta \tag{63}$$

With these gains, the Lyapunov derivative satisfies:

$$\dot{V}_\mathfrak{z} \leq -\xi_1|e_1|^{3/2} - (\xi_2 - \Delta)|e_2| \tag{64}$$

This inequality proves that $\dot{V}_\mathfrak{z}$ is negative definite, which implies that the estimation errors e_1 and e_2 converge to zero in finite time.

Remark 6. Lyapunov stability analysis proves that FOST-SMD is finite-time stable. Moreover, estimation errors e_1 and e_2 converge to zero in a finite time to provide precise and robust derivative estimation. The choice of gains ξ_1 and ξ_2 is critical for ensuring stability and performance.

This design provides a robust and accurate solution for derivative estimation in continuous and discrete time.

The numerical simulations presented in the subsequent section further validate the proposed differentiator for state estimation and control performance.

4. Simulation results

Numerical simulations are carried out in the MATLAB/Simulink environment in order to validate the efficiency and superiority of the proposed control algorithm in terms of trajectory tracking precision, robustness against bounded uncertainties, and noise tolerance. The velocity states of underactuated subsystems are estimated along with nonlinear drift functions by FOSTSMD and DRNAM, with and without FOSTSMD, respectively (Algorithm 1).

The parameters of the quadcopter model used in the simulations are given as: mass $m = 0.650$ kg, inertia coefficients $I_x = I_y = I_z = 7.5\times10^{-3}$ kg.m^2, thrust coefficient $b = 3.13\times10^{-5}$ N.s^2, drag coefficient $d = 7.5\times10^{-7}$ N.m.s^2, rotor inertia coefficient $I_r = 6\times10^{-5}$ kg.m^2, and arm length

Algorithm 1 Fractional-Order Super-Twisting Sliding Mode Differentiator.

Require: Measured signal $\mathfrak{z}_1(t)$, order $\alpha \in (0, 1)$, design gains $\xi_1 > 0$, $\xi_2 > \Delta$, sampling time T_s

Ensure: Estimated derivative $\hat{\mathfrak{z}}_2(t) \approx D^\alpha \mathfrak{z}_1(t)$

Initialize: $\hat{\mathfrak{z}}_1(0) \leftarrow \mathfrak{z}_1(0)$, $\hat{\mathfrak{z}}_2(0) \leftarrow 0$, $t \leftarrow 0$

while $t \leq T_{\text{final}}$ **do**

  Compute estimation error: $e_1(t) \leftarrow \mathfrak{z}_1(t) - \hat{\mathfrak{z}}_1(t)$

  Update estimate of $\dot{\hat{\mathfrak{z}}}_1(t)$:

$$\dot{\hat{\mathfrak{z}}}_1(t) \leftarrow \xi_1 \sqrt{|e_1(t)|} \cdot \; sign\,(e_1(t)) + \hat{\mathfrak{z}}_2(t)$$

  Compute fractional-order derivative:

$$D^\alpha \hat{\mathfrak{z}}_2(t) \leftarrow \xi_2 \cdot \; sign\,(e_1(t))$$

  ▷ Use Caputo or Grünwald–Letnikov approximation

  Update state estimates:

$$\hat{\mathfrak{z}}_1(t + T_s) \leftarrow \hat{\mathfrak{z}}_1(t) + T_s \cdot \dot{\hat{\mathfrak{z}}}_1(t)$$

$$\hat{\mathfrak{z}}_2(t + T_s) \leftarrow \text{FractionalIntegrator}(D^\alpha \hat{\mathfrak{z}}_2(t))$$

  Increment time: $t \leftarrow t + T_s$

end while

return $\hat{\mathfrak{z}}_1(t), \hat{\mathfrak{z}}_2(t)$

$l = 0.23$ m. Matched uncertainties are imposed on the system modeled as $\Delta = K_\Delta \sin(2\pi t)$. The initial translational and angular positions are chosen as [0 0 0] m and [0 0 0] rad, respectively. The reference trajectory in the x, y, and z directions is generated through the use of the Simulink Signal Builder. Finally, observer and controller gains are tuned by trial and error to make a trade-off between the amplitude of the control inputs and the convergence speed. For the numerical results, the control parameters $(\alpha_\mathfrak{z}, \beta_\mathfrak{z}, \gamma_\mathfrak{z}, \kappa_1, \kappa_2, \xi_1, \xi_2, \eta_F, \eta_H)$ were systematically selected according to the Lyapunov-based stability conditions and finite-time convergence criteria derived in the manuscript. Specifically, $\alpha_\mathfrak{z} = 0.8$, $\beta_\mathfrak{z} = 1.2$, and $\gamma_\mathfrak{z} = 0.6$ were chosen to balance memory effects and transient response in the fractional sliding surface. The super-twisting gains $\kappa_1 = 3.5$ and $\kappa_2 = 2.0$ ensured finite-time convergence while suppressing chattering. Neural learning rates were set as $\xi_1 = 0.01$ and $\xi_2 = 0.005$ to achieve rapid yet stable adaptation, while regularization parameters $\eta_F = 0.001$ and $\eta_H = 0.001$ prevented overfitting and improved generalization in the DRNAM estimator. Similarly, subsystem-specific gains for translational and rotational channels were selected as $(x, y, \varphi, \vartheta, \psi) = (3.0, 3.0, 2.5, 2.5, 2.0)$, guaranteeing robust trajectory tracking with bounded approximation errors and a balanced trade-off between convergence speed and disturbance rejection. These values were not arbitrarily tuned but carefully derived to ensure negative definiteness of the Lyapunov function and strong robustness against uncertainties.

Let us also note that the selected reference trajectory was intentionally constructed to include multiple validation scenarios in a single test. In particular, it contains step changes in altitude and lateral position (step response), turns and quick changes (aggressive maneuvers), and portions with instantaneous external perturbations, such as wind gusts (disturbance rejection). Such embedded scenarios enable full validation of the robustness and adaptability of the proposed NA-FOSTSMC framework, with no separate test cases required.

Furthermore, to provide confirmation of the superior capability of the proposed approach, a comparative analysis is performed in the presence and absence of FOSTSMD and DRNAM, which have a strong track record of performance. Figs. 4–6 illustrate the tracking performance of the proposed flight controller and show that the proposed controller allows fast finite-time convergence while maintaining robustness against matched uncertainties. Similarly, Figs. 7–9 present the angular regulation response of the proposed control law with and without FOSTSMD and DRNAM. It has been possible for the proposed control algorithm

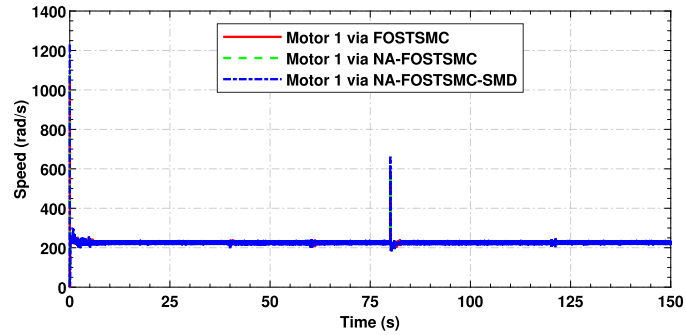

Fig. 16. Time history of control input ω_1 with NA-FOSTSMC, to show the behavior of the system in dynamic case.

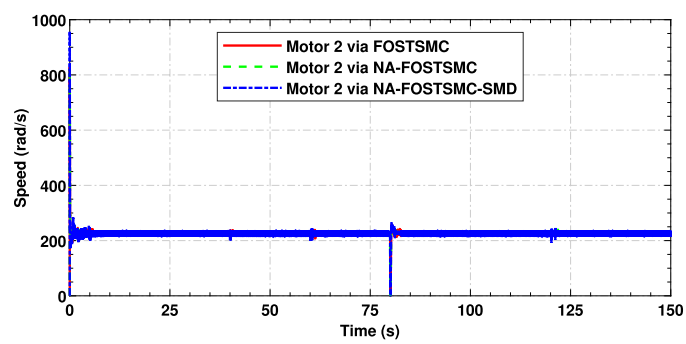

Fig. 17. Time history of control input ω_2, which shows the control effort produced by the NA-FOSTSMC algorithm to obtain accurate trajectory tracking along with system stability.

to successfully drive the actual positions and orientations of the quadcopter to follow the reference trajectories in finite time, despite the sudden variations in desired translational and rotational positions at each instant. A more precise and encompassing comparison is provided by the quantitative performance evaluation given in Table 1, based on key attributes: namely, steady-state errors, maximum overshoot percentage, and settling time.

The estimates for the translational and angular subsystem velocity states are depicted in Figs. 10–12 and 13–15, respectively. It can be clearly seen that the estimated values track the desired velocities without much deviation, thus showing the effectiveness of the proposed FOSTSMD observer in rejecting external disturbances. Further, the tracking results confirm that the proposed control algorithm is practically applicable, as it requires only limited state information of the quadcopter. More importantly, its robust behaviour against noisy measurements and partial sensor failures guarantees the superiority of FOSTSMD and makes the proposed controller a very good candidate in the existing literature. With reduced dependence on sensor measurements, the chances of sensor failure are greatly diminished, thus guaranteeing reliability and stability for the controller in real-world applications.

Time histories of applied control inputs $\omega_1, \omega_2, \omega_3$, and ω_4 under NA-FOSTSMC law are given in Fig. 16–19. It can be clearly seen that the nature of control inputs is smooth while their magnitudes are within feasible limits. In addition, the output tracking error is ensured to converge to zero in finite time, and the external disturbance can also be attenuated efficiently by the suggested controller. Interestingly, the NA-FOSTSMC law is free of high-frequency switching, unlike the conventional FOSTSM law, which exhibits chattering-an undesirable phenomenon in practical applications. Actually, the present algorithm is very effective in reducing chattering and is therefore well applicable in practical applications.

The performances of the subsystems obtained by the proposed method are better at least in one performance index in Tables 1 and

Table 1
Comparison of quantitative performance measures over controllers for subsystems. The first column shows the metrics, and the controller results are aligned in columns for better readability..

Metric	Subsystem	FOSTSMC	NA-FOSTSMC	NA-FOSTSMC-SMD
Max. Overshoot [%]	ψ	0.0628	0.0452	0.0362
	z	0.0128	0.0179	0.0165
	y	0.2371	0.2256	0.2389
	x	0.9379	0.8934	0.7563
Steady-State Error	ψ	0.000837	0.000345	0.000297
	z	0.00263	0.00234	0.00156
	y	0.00826	0.00782	0.00623
	x	0.00671	0.00621	0.00432
Settling Time (s)	ψ	5.5296	6.2173	4.2583
	z	2.2845	2.1326	1.9273
	y	6.4715	6.0284	6.1263
	x	3.5629	3.0326	3.0083
Integral of Absolute Error (IAE)	ψ	0.1284	0.0923	0.0712
	z	0.0845	0.0678	0.0526
	y	0.2632	0.2156	0.1987
	x	0.3421	0.2987	0.2435
Integral of Squared Control Input (ISCI)	ψ	1.823	1.452	1.236
	z	1.276	1.084	0.932
	y	2.315	1.987	1.743
	x	2.874	2.423	2.018

Table 2
Qualitative analysis of robustness, adaptation, model reliance, chattering suppression, velocity estimation accuracy and convergence rate conventional controllers among the proposed NA-FOSTSMC..

Attribute	PID	Classical SMC	Standard ST-SMC	Proposed NA-FOSTSMC
Robustness to disturbances	Low	High	High	**Very High**
Adaptability to nonlinearities	Low	Moderate	Moderate	**High (via DRNAM)**
Dependence on accurate model	High	Moderate	Moderate	**Low (learning-based)**
Chattering alleviation	N/A	Poor	Moderate	**Excellent (fractional-order ST)**
Velocity estimation under noise	Poor	Poor	Moderate	**Accurate (FOST-D)**
Convergence speed	Slow	Moderate	Fast	**Finite-time, fastest**

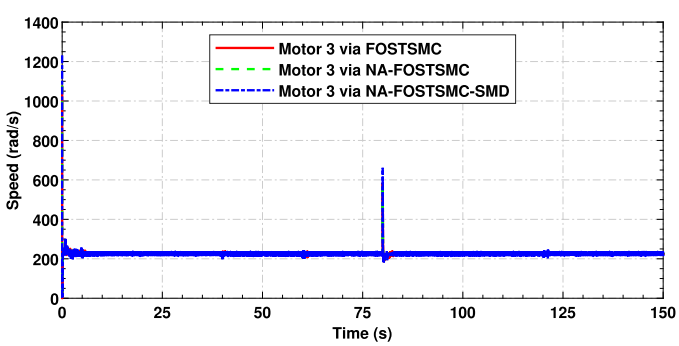

Fig. 18. Time history of control input ω_3, illustrating the control effort generated by the NA-FOSTSMC scheme to achieve precise trajectory tracking and system stability.

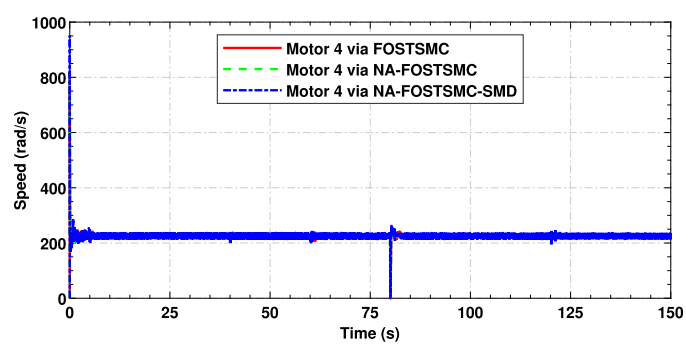

Fig. 19. Time history of control input ω_4, depicting the exerted control effort by the NA-FOSTSMC method to realize accurate trajectory following and system stability.

2 for all subsystems. In terms of numerical performance, it achieves the smallest overshoot, steady-state error, and settling time, and greatly reduces the Integral of Absolute Error (IAE) and Integral of Squared Control Input (ISCI) values, confirming the best tracking accuracy with the least control effort. Its practical high performance, robustness, and convergence speed are illustrated, indicating the effectiveness of the proposed methodology compared with other traditional controllers. To justify the advantages of the NA-FOSTSMC scheme, a qualitative comparison is made with traditional control methods, including PID, classical SMC, and standard ST-SMC. PID controllers, being simple and ubiquitous [50], are known to provide weak robustness in nonlinear dynamics and disturbances. Classical SMC improves the robustness but propagates severe chattering and relies on exact modeling [51], while the conventional ST-SMC reduces chattering and increases the convergence rate, it is still relying on full accurate system information and can not handle with noise corrupted measurements [52].

On the other hand, the NA-FOSTSMC is developed by incorporating fractional-order calculus, DRNAM learning, and FOST-D differentiators, which bring finite-time convergence, high disturbance rejection, and trustworthy velocity estimation in the presence of noise. This architecture provides greater flexibility and robustness than any traditional controller can, thereby validating the superiority aspired to by the proposed method.

In practice, both NA-FOSTSMC and NA-FOSTSMC-SMD controllers are realizable in real time, since the fractional-order sliding mode and differentiator are very simple from a computational load perspective.

The solution of introducing deep residual neural networks with attention introduces higher computational complexity than classical controllers, but can be handled with GPU/FPGA acceleration on modern embedded platforms.

5. Conclusion

This paper proposed a fractional-order super-twisting sliding mode control for quadcopter trajectory tracking under uncertainties. By reformulating the system into an input-decoupled form and integrating fractional calculus with super-twisting design, finite-time convergence and chattering suppression were achieved. A deep residual neural network with attention mechanisms estimated unknown nonlinearities, reducing model dependence, while a fractional-order super-twisting differentiator ensured accurate velocity estimation from position data. Analytical stability was proven via Lyapunov theory, with tracking errors converging to bounded regions. Quantitative metrics (IAE, ISCI) also verified the reduction in error and control magnitude, and the simulations showed faster convergence, a smaller steady-state error, and less overshoot compared with traditional methods. Though there are still limitations in computing complexity, data dependency, and parameter tuning, future research will focus on developing lightweight models, adaptive strategies, and hardware-in-the-loop verification to improve real-time applicability.

CRediT authorship contribution statement

Safeer Ullah: Writing – review & editing, Writing – original draft, Supervision, Software, Methodology, Formal analysis, Conceptualization; **Yukang Cui:** Writing – review & editing, Investigation, Formal analysis; **Hisham Alghamdi:** Writing – review & editing, Project administration, Investigation, Conceptualization; **Sultan Alghamdi:** Validation, Project administration, Data curation; **Ahmed S. Alsafran:** Writing – review & editing, Validation, Project administration, Investigation, Conceptualization; **Abdullah A. Algethami:** Writing – review & editing, Validation, Investigation; **Dong-Won Jung:** Writing – review & editing, Project administration, Funding acquisition.

Data availability

No data was used for the research described in the article.

Declaration of competing interest

The authors declare no conflicts of interest related to this work. The research was conducted independently, without any financial or personal relationships that could have influenced the study. All authors have approved the final version of the manuscript and agree with its submission to Aerospace Science and Technology.

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References

[1] O. Mechali, L. Xu, X. Xie, J. Iqbal, Theory and practice for autonomous formation flight of quadrotors via distributed robust sliding mode control protocol with fixed-time stability guarantee, Control Eng. Pract. 123 (2022) 105150.
[2] L.A. Al-Haddad, A.A. Jaber, W. Giernacki, Z.H. Khan, K.M. Ali, M.A. Tawafik, A.J. Humaidi, Quadcopter unmanned aerial vehicle structural design using an integrated approach of topology optimization and additive manufacturing, Designs 8 (3) (2024) 58.
[3] E.C.-Y. Yang, P.C.-P. Chao, C.-K. Sung, Optimal control of an under-actuated system for landing with desired postures, IEEE Trans. Control Syst. Technol. 19 (2) (2011) 248–255.
[4] E.C.-Y. Yang, P.C.-P. Chao, C.-K. Sung, Optimal control of an under-actuated system for landing with desired postures, IEEE Trans. Control Syst. Technol. 19 (2) (2010) 248–255.
[5] C. Souza, G.V. Raffo, E.B. Castelan, Passivity based control of a quadrotor UAV, IFAC Proceed. Vol. 47 (3) (2014) 3196–3201.
[6] S. Yang, B. Xian, Energy-based nonlinear adaptive control design for the quadrotor UAV system with a suspended payload, IEEE Trans. Ind. Electron. 67 (3) (2019) 2054–2064.
[7] Z. Jia, J. Yu, Y. Mei, Y. Chen, Y. Shen, X. Ai, Integral backstepping sliding mode control for quadrotor helicopter under external uncertain disturbances, Aerosp. Sci. Technol. 68 (2017) 299–307.
[8] S. Ullah, A. Mehmood, Q. Khan, S. Rehman, J. Iqbal, Robust integral sliding mode control design for stability enhancement of under-actuated quadcopter, Int. J. Control Autom. Syst. (2020) 1–08.
[9] S. Ullah, Q. Khan, A. Mehmood, R. Akmeliawati, Integral backstepping based robust integral sliding mode control of underactuated nonlinear electromechanical systems, J. Control Eng. Appl. Inf. 21 (3) (2019) 42–50.
[10] K. Elyaalaoui, M. Labbadi, M. Ouassaid, M. Cherkaoui, M. Bouzi, High-order sliding-mode control using integral terminal sliding manifold for a wind turbine under grid faults, Int. J. Control 97 (3) (2024) 635–650.
[11] N.B. Abera, C.M. Abdissa, L.N. Lemma, An improved nonsingular adaptive super twisting sliding mode controller for quadcopter, PLoS ONE 19 (10) (2024) e0309098. https://doi.org/10.1371/journal.pone.0309098
[12] K. Ichida, K. Watanabe, K. Izumi, N. Uchida, Fuzzy switching control of underactuated manipulators with approximated switching regions, in: Intelligent Robots and Systems, 2006 IEEE/RSJ International Conference on, IEEE, 2006, pp. 586–591.
[13] R.-J. Wai, M.-A. Kuo, J.-D. Lee, Cascade direct adaptive fuzzy control design for a nonlinear two-axis inverted-pendulum servomechanism, IEEE Trans. Syst., Man, Cybernet., Part B (Cybernetics) 38 (2) (2008) 439–454.
[14] C. Zhu, J. Chen, M. Iwasaki, H. Zhang, Event-triggered deep learning control of quadrotors for trajectory tracking, IEEE Trans. Ind. Electron. 71 (3) (2023) 2726–2736.
[15] D. Jang, C.Y. Son, J. Yoo, H.J. Kim, K.H. Johansson, Efficient networked UAV control using event-triggered predictive control, IFAC-PapersOnLine 52 (15) (2019) 412–417.
[16] F. Santoso, M.A. Garratt, S.G. Anavatti, I. Petersen, Robust hybrid nonlinear control systems for the dynamics of a quadcopter drone, IEEE Trans. Syst., Man, and Cybernet.: Systems 50 (8) (2018) 3059–3071.
[17] Y. Guo, L. Luo, C. Bao, Design of a fixed-Wing UAV controller combined fuzzy adaptive method and sliding mode control, Math. Problems Eng. 2022 (1) (2022) 2812671.
[18] Y. Liang, J. Yang, L. Zhang, S. Baldi, B. De Schutter, Switched control design for quadrotor in target tracking with complex intermittent measurements, J. Guidance, Control Dynamics 46 (1) (2023) 206–214.
[19] C.P. E.Y. Huanca, G.P. Incremona, P. Colaneri, Design of a distributed switching model predictive control for quadrotor UAVs aggregation, IEEE Control Systems Lett. 7 (2023) 2964–2969.
[20] K.D. Do, Global tracking control of underactuated ODINs in three-dimensional space, Int. J. Control 86 (2) (2013) 183–196.
[21] R. Cui, J. Guo, Z. Mao, Adaptive backstepping control of wheeled inverted pendulums models, Nonlinear Dyn. 79 (1) (2015) 501–511.
[22] Y. Bai, C. Zhang, Finite-time tracking control of a quadrotor UAV without velocity measurement, in: 2017 2Nd International Conference on Robotics and Automation Engineering (ICRAE), IEEE, 2017, pp. 234–238.
[23] U. Javaid, A. Mehmood, J. Iqbal, A.A. Uppal, Neural network and URED observer based fast terminal integral sliding mode control for energy efficient polymer electrolyte membrane fuel cell used in vehicular technologies, Energy 269 (2023) 126717.
[24] M. Guisser, H. Medromi, A high gain observer and sliding mode controller for an autonomous quadrotor helicopter, Int. J. Intell. Control Syst. 14 (3) (2009) 204–212.
[25] W. Cai, J. She, M. Wu, Y. Ohyama, Disturbance suppression for quadrotor UAV using sliding-mode-observer-based equivalent-input-disturbance approach, ISA Trans. (2019).
[26] A. Levant, Principles of 2-sliding mode design, Automatica 43 (4) (2007) 576–586.
[27] A. Bayrak, M.Ö. Efe, Disturbance observer-based control schemes for quadrotors–a tutorial, Int. J. Gen. Syst. 51 (3) (2022) 215–238.
[28] Y. Cheng, L. Jiang, T. Li, L. Guo, Robust tracking control for a quadrotor UAV via DOBC approach, in: 2018 Chinese Control and Decision Conference (CCDC), IEEE, 2018, pp. 559–563.
[29] R. Yuan, X. Tan, G. Fan, J. Yi, Robust adaptive neural network control for a class of uncertain nonlinear systems with actuator amplitude and rate saturations, Neurocomputing 125 (2014) 72–80.
[30] H. Mo, G. Farid, Nonlinear and adaptive intelligent control techniques for quadrotor UAV–a survey, Asian J. Control. 21 (2) (2019) 989–1008.
[31] W. Koch, R. Mancuso, R. West, A. Bestavros, Reinforcement learning for UAV attitude control, ACM Trans. Cyber-Phys. Syst. 3 (2) (2019) 1–21.
[32] B. Ma, Z. Liu, Q. Dang, W. Zhao, J. Wang, Y. Cheng, Z. Yuan, Deep reinforcement learning of UAV tracking control under wind disturbances environments, IEEE Trans. Instrum. Meas. 72 (2023) 1–13.
[33] A.F. Abdul Ghaffar, T. Richardson, C. Greatwood, A combined model reference adaptive control law for multirotor UAVs, IET Control Theory Appl. 15 (11) (2021) 1474–1487.

[34] O. Akgün, E. Subaşı, T. Türker, A lyapunov based model reference adaptive control of a quadrotor, in: 2017 10Th International Conference on Electrical and Electronics Engineering (ELECO), IEEE, 2017, pp. 732–736.
[35] A. Hamza, A.H. Mohamed, A. El-Badawy, Robust H-infinity control for a quadrotor UAV, in: AIAA SCITECH 2022 Forum, 2022, p. 2033.
[36] S. Panza, M. Lovera, M. Sato, K. Muraoka, Structured μ-synthesis of robust attitude control laws for quad-tilt-wing unmanned aerial vehicle, J. Guidance, Control, Dynam. 43 (12) (2020) 2258–2274.
[37] Z. Cai, S. Zhang, X. Jing, Model predictive controller for quadcopter trajectory tracking based on feedback linearization, IEEE Access 9 (2021) 162909–162918.
[38] A. Eskandarpour, M. Mehrandezh, K. Gupta, A. Ramirez-Serrano, M. Soltanshah, A constrained robust switching MPC structure for tilt-rotor UAV trajectory tracking problem, Nonlinear Dyn. 111 (18) (2023) 17247–17275.
[39] H. Merabti, I. Bouchachi, K. Belarbi, Nonlinear model predictive control of quadcopter, in: 2015 16Th International Conference on Sciences and Techniques of Automatic Control and Computer Engineering (STA), IEEE, 2015, pp. 208–211.
[40] D. Chen, L. Xu, C. Wang, An advanced control method for aircraft carrier landing of UAV based on CAPF–NMPC, Aerospace 11 (8) (2024) 656.
[41] J. Espinoza, N. Hakim, D. Tan, T. Wilson, K. Bingi, E. Khan, S. Masrura, Fractional-order pid control of quadrotor drone, in: 2023 Innovations in Power and Advanced Computing Technologies (I-PACT), IEEE, 2023, pp. 1–6.
[42] A.-W.A. Saif, K.B. Gaufan, S. El-Ferik, M. Al-Dhaifallah, Fractional order sliding mode control of quadrotor based on fractional order model, IEEE Access 11 (2023) 79823–79837.
[43] S. Ullah, Q. Khan, A. Mehmood, S.A.M. Kirmani, O. Mechali, Neuro-adaptive fast integral terminal sliding mode control design with variable gain robust exact differentiator for under-actuated quadcopter UAV, ISA Trans. 120 (2022) 293–304.
[44] Z. Zhang, H. Zhang, Fractional-Order sliding mode with active disturbance rejection control for UAVs, Applied Sci. 15 (2) (2025) 556. https://doi.org/10.3390/app15020556
[45] E.W. Metekia, W.A. Asfaw, C.M. Abdissa, L.N. Lemma, Control of a fixed wing unmanned aerial vehicle using a robust fractional order controller, Sci. Rep. 14 (2024) 12345. https://doi.org/10.1038/s41598-024-12345-y
[46] C. Li, Y. Chen, Stability of fractional-order nonlinear dynamic systems: lyapunov direct method and generalized mittag-Leffler stability, Math. Comput. Simul. 82 (5) (2011) 851–859.
[47] C.A. Monje, Y. Chen, B. Vinagre, D. Xue, V. Feliu, Fractional-order Systems and Controls: Fundamentals and Applications, Springer, 2010.
[48] I. Podlubny, Fractional Differential Equations, Academic Press, 1999.
[49] R.K. Sharma, X. Xiong, S. Kamal, S. Ghosh, Discrete-time super-twisting fractional-order differentiator with implicit euler method, IEEE Trans. Circuits Syst. II Express Briefs 68 (4) (2020) 1238–1242.
[50] C. Van Nguyen, M.T. Nguyen, Design and simulation of PID-Based control system for UAV quadcopters, in: International Conference on Advances in Information and Communication Technology, 848, Springer, 2023, pp. 146–153.
[51] A. Yesmin, A. Sinha, Sliding mode controller for quadcopter UAVs: a comprehensive survey, Drones 9 (9) (2025) 625. https://doi.org/10.3390/drones9090625
[52] Y. Ye, S. Hu, X. Zhu, Z. Sun, An improved super-Twisting sliding mode composite control for quadcopter UAV formation, Machines 12 (1) (2024) 32. https://doi.org/10.3390/machines12010032